LAB EXPERIMENTS

Day 4-DWDM

1.Consider that you are owning a supermarket mall and through membership cards, you have some basic data about your customers like Customer ID, age, gender, annual income and spending score. For the above scenario, the Problem Statement was You want to understand the customers who can easily converge [Target Customers] so that the data can be given to the marketing team and plan the strategy accordingly. For the above scenario prepare a dataset and perform **Clustering Analysis** to segment the customers in the Mall. There are clearly Five segments of Customers based on their Annual Income and Spending Score namely *Usual Customers, Priority Customers, Senior Citizen Target Customers, and Young Target Customers.*Sample data

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Experiment 1: Mall Customer Clustering Using K-Means in WEKA

Aim

To perform K-Means clustering on mall customer data using WEKA and group customers into five clusters based on their annual income and spending score.

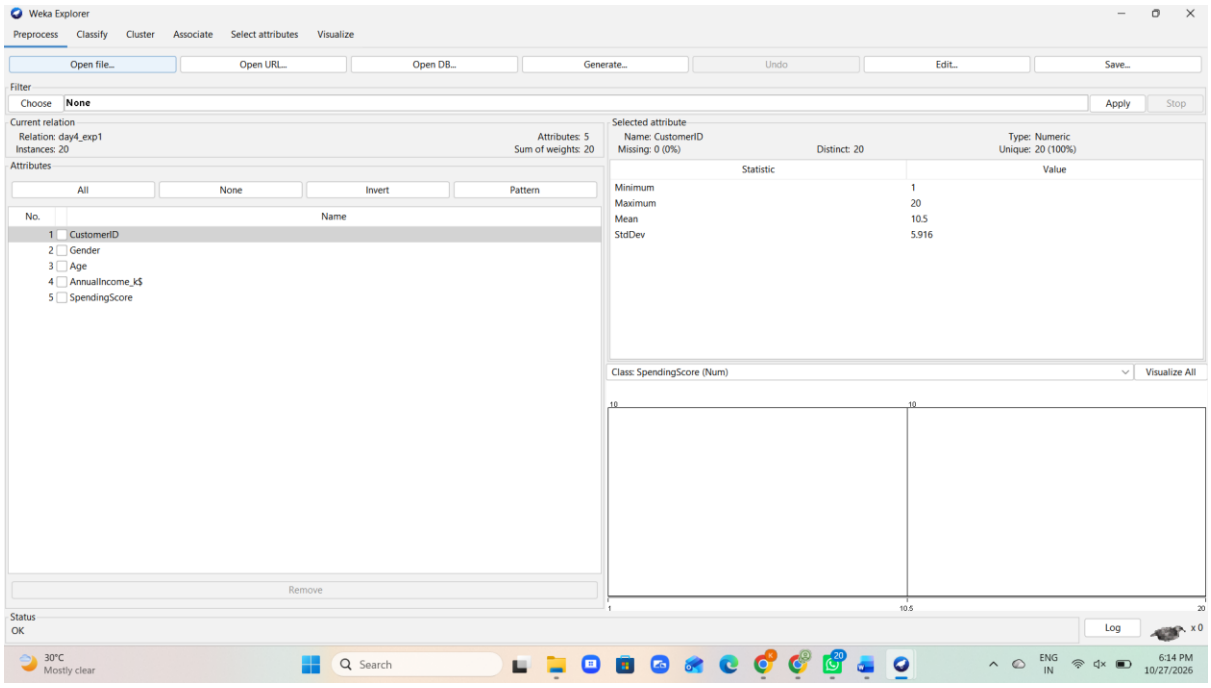
Procedure

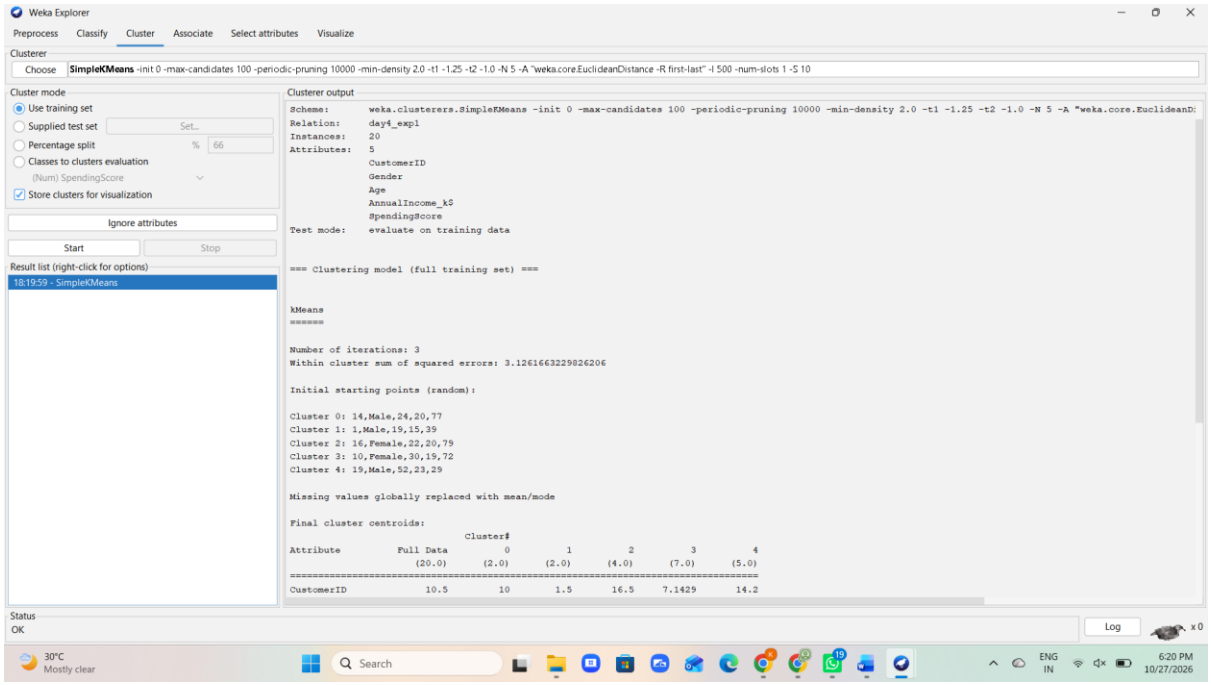
1. Open WEKA Explorer and select the Preprocess tab.
2. Click Open file and load the day4_exp1.csv dataset.
3. Click the Cluster tab and select SimpleKMeans.
4. Configure the number of clusters (numClusters) to 5.
5. Select Use training set and enable Store clusters for visualization.
6. Click Start to perform the clustering process.
7. Observe the cluster centroids and the number of instances assigned to each cluster.
8. Right-click the clustering result and select Visualize cluster assignments.

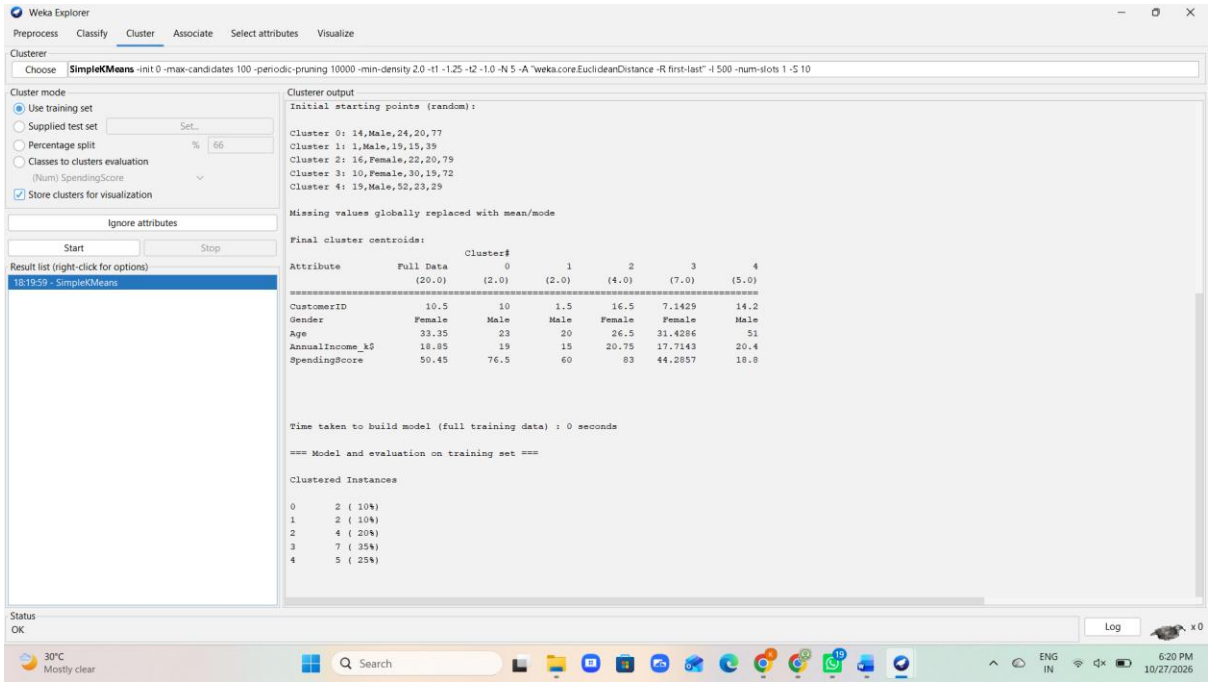
9. Set the X-axis to AnnualIncome_k$, the Y-axis to SpendingScore, and the colour to Cluster.

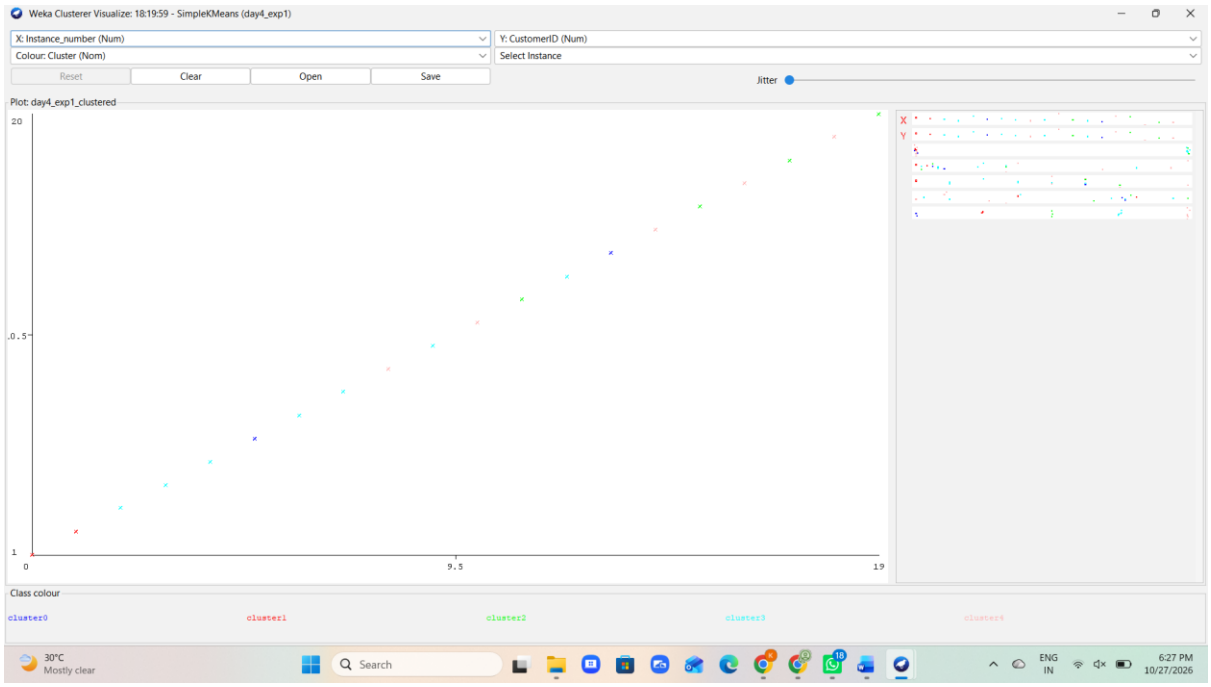
10. Save the graph for the experiment report.

Figure 1: Mall Customer Segmentation Using K-Means Clustering in WEKA.









Result

K-Means clustering was successfully performed in WEKA, and the 20 customer records were grouped into five clusters. The largest cluster contained 7 customers (35%), while the smallest clusters contained 2 customers each (10%). The cluster visualization displays the customer groups in different colours.

2.Create the following dataset using CSV file format. To perform cluster analysis using K-Means in WEKA. To change the cluster size and plot the graph and illustrate the visualization of cluster.

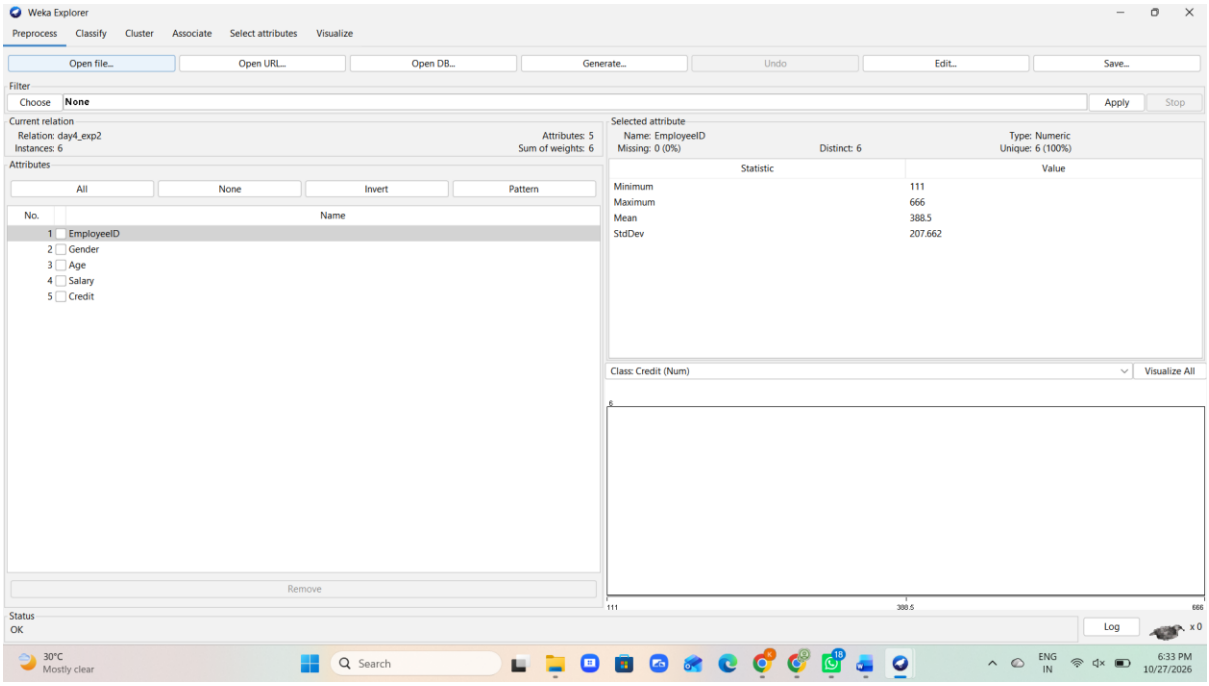
EmployeID	Gender	Age	Salary	Credit
111	Male	28	150000	39
222	Male	25	150000	27
333	Female	26	160000	42
444	Female	25	160000	40
555	Female	30	170000	64
666	Male	29	200000	72

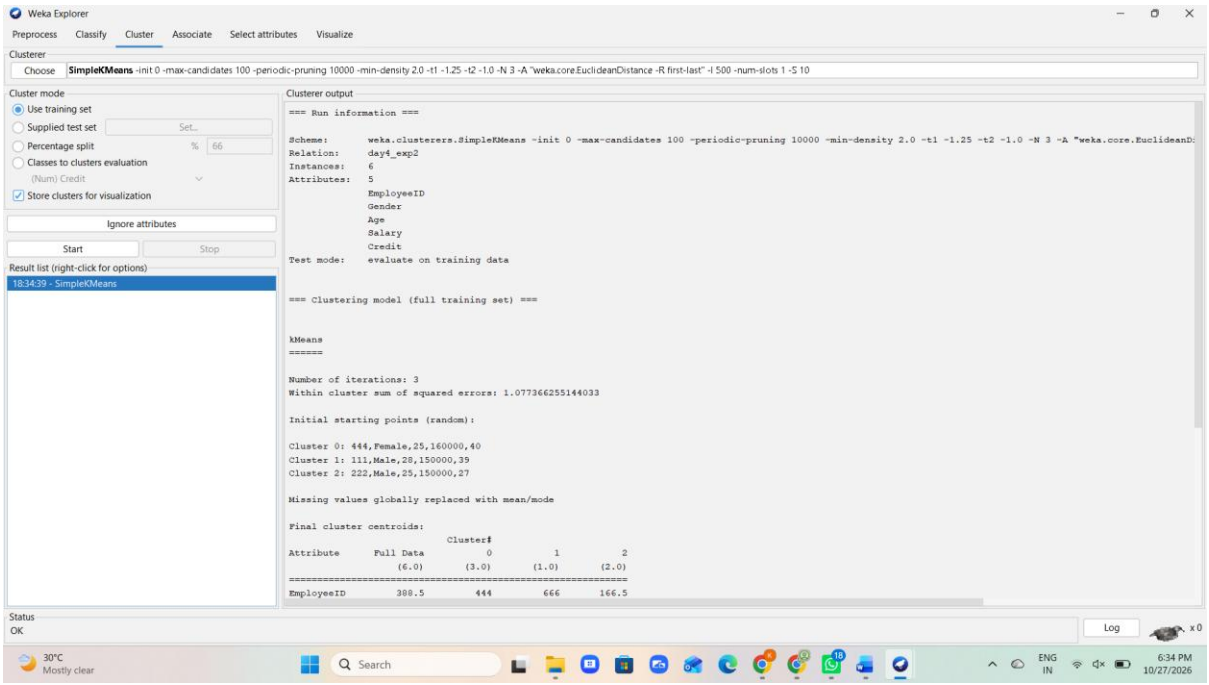
Aim

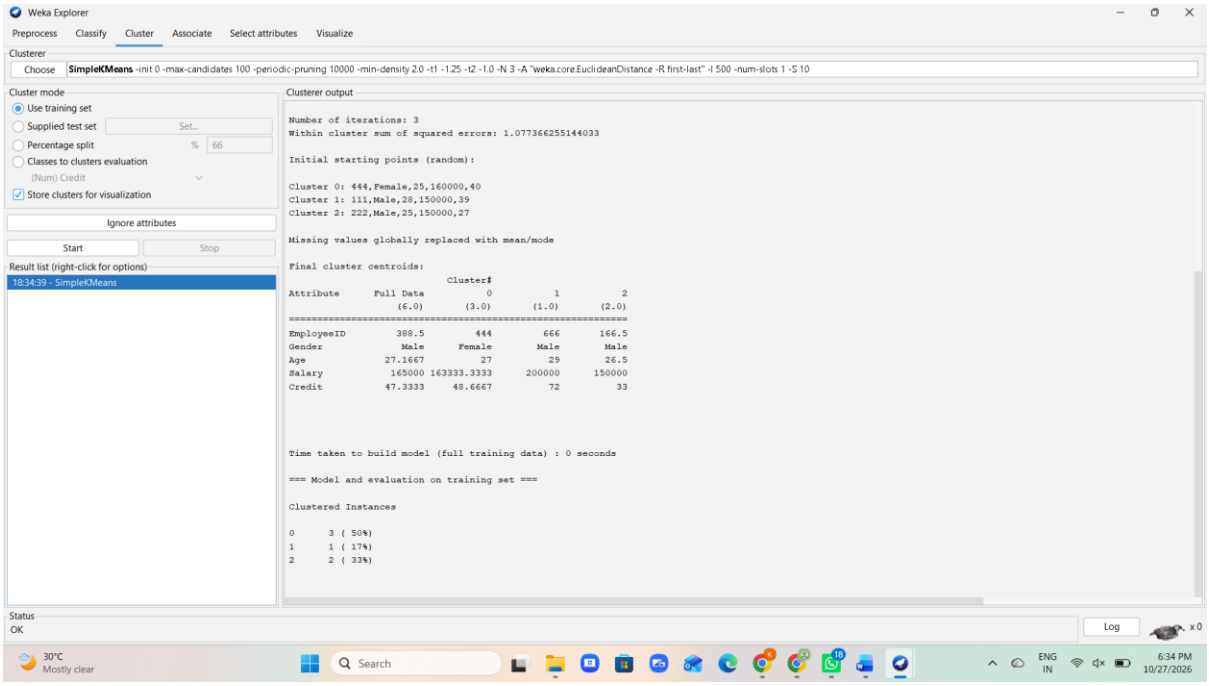
To perform cluster analysis using the K-Means algorithm in WEKA, vary the number of clusters, and visualize the clusters using the given employee dataset.

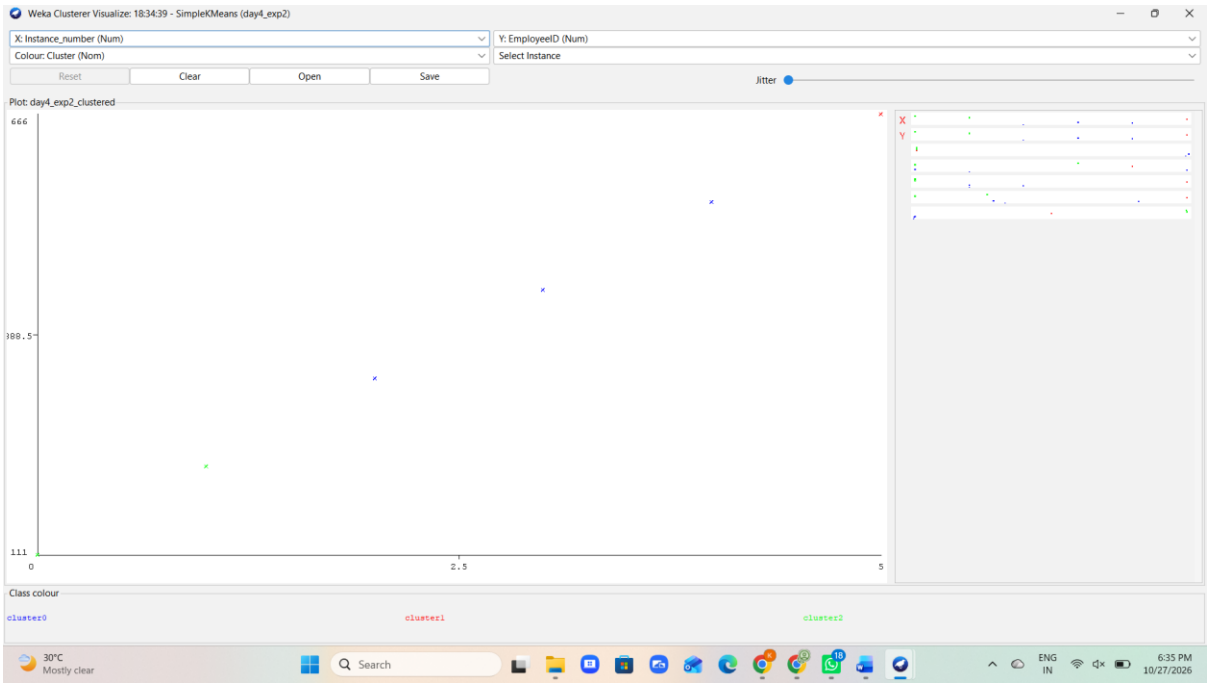
Procedure

1. Open WEKA Explorer and select the Preprocess tab.
2. Click Open file and load day4_exp2.csv.
3. Click the Cluster tab and select SimpleKMeans.
4. Set the number of clusters (numClusters) to 2.
5. Select Use training set and enable Store clusters for visualization.
6. Click Start to execute the clustering algorithm.
7. Record the cluster assignments and the number of employees in each cluster.
8. Repeat the process with 3 and 4 clusters by changing the numClusters value.
9. Right-click each result and select Visualize cluster assignments.
10. Set the X-axis to Age, the Y-axis to Credit, and the colour to Cluster.
11. Save the graphs and compare the clustering results.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: day4_exp2
Instances: 6
Attributes: 5
Sum of weights: 6
Selected attribute
Name: EmployeeID
Missing: 0 (0%)
Distinct: 6
Type: Numeric
Unique: 6 (100%)
Attributes
All
None
Invert
Pattern
No.
Name
1 EmployeeID
2 Gender
3 Age
4 Salary
5 Credit
Statistic
Value
Minimum 111
Maximum 666
Mean 388.5
StdDev 207.662
Class: Credit (Num)
Visualize All
Remove
Status
OK
Log
x 0
30°C
Mostly clear
Search
ENG
IN
6:33 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
%
66
Classes to clusters evaluation
(Num) Credit
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
18:34:39 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanD
Relation: day4_exp2
Instances: 6
Attributes: 5
EmployeeID
Gender
Age
Salary
Credit
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 3
Within cluster sum of squared errors: 1.077366255144033
Initial starting points (random):
Cluster 0: 444,Female,25,160000,40
Cluster 1: 111,Male,28,150000,39
Cluster 2: 222,Male,25,150000,27
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1 2
(6.0) (3.0) (1.0) (2.0)
EmployeeID 388.5 444 666 166.5
Status
OK
Log
x 0
30°C
Mostly clear
Search
ENG
IN
6:34 PM
10/27/2026





Result

K-Means clustering was performed on the employee dataset using WEKA. The number of clusters was varied to 2, 3, and 4, and the results were visualized using scatter plots of Age versus Credit. The visualizations demonstrate how employees are grouped differently when the number of clusters changes.

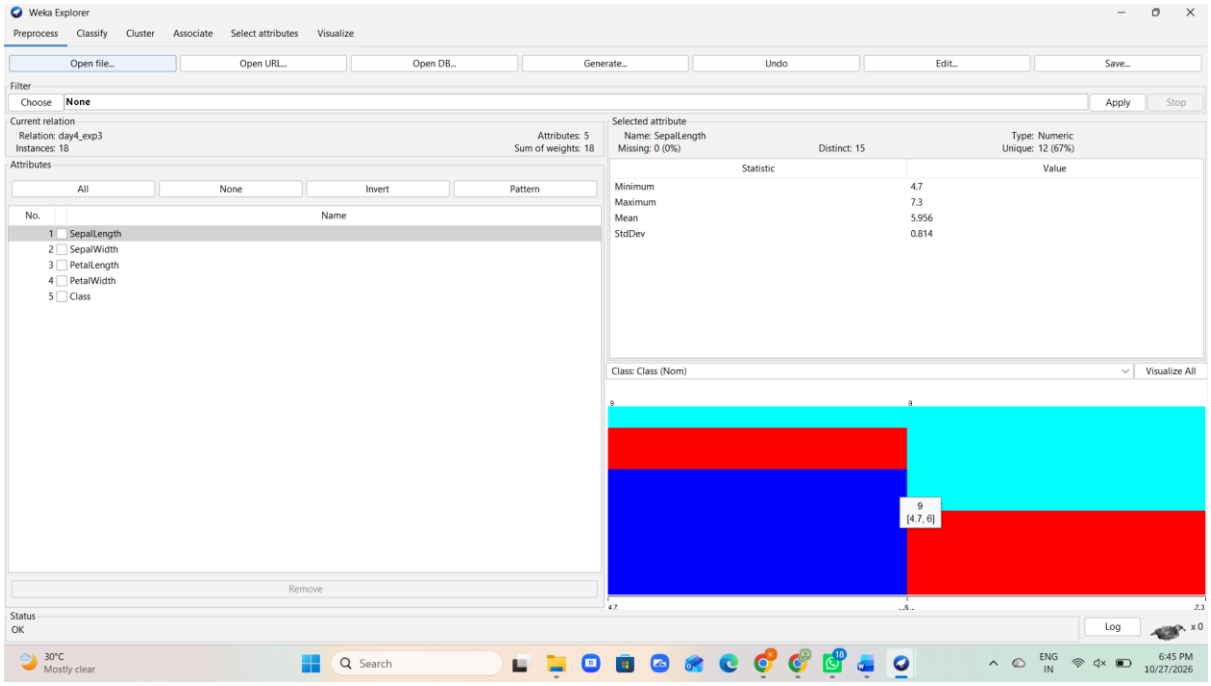
3.Prediction of categorical data using Naïve Bayes classification through WEKA using any datasets. Compare the Naïve Bayes algorithm with SVM using the summary of results given by the classifiers and plot the graph.

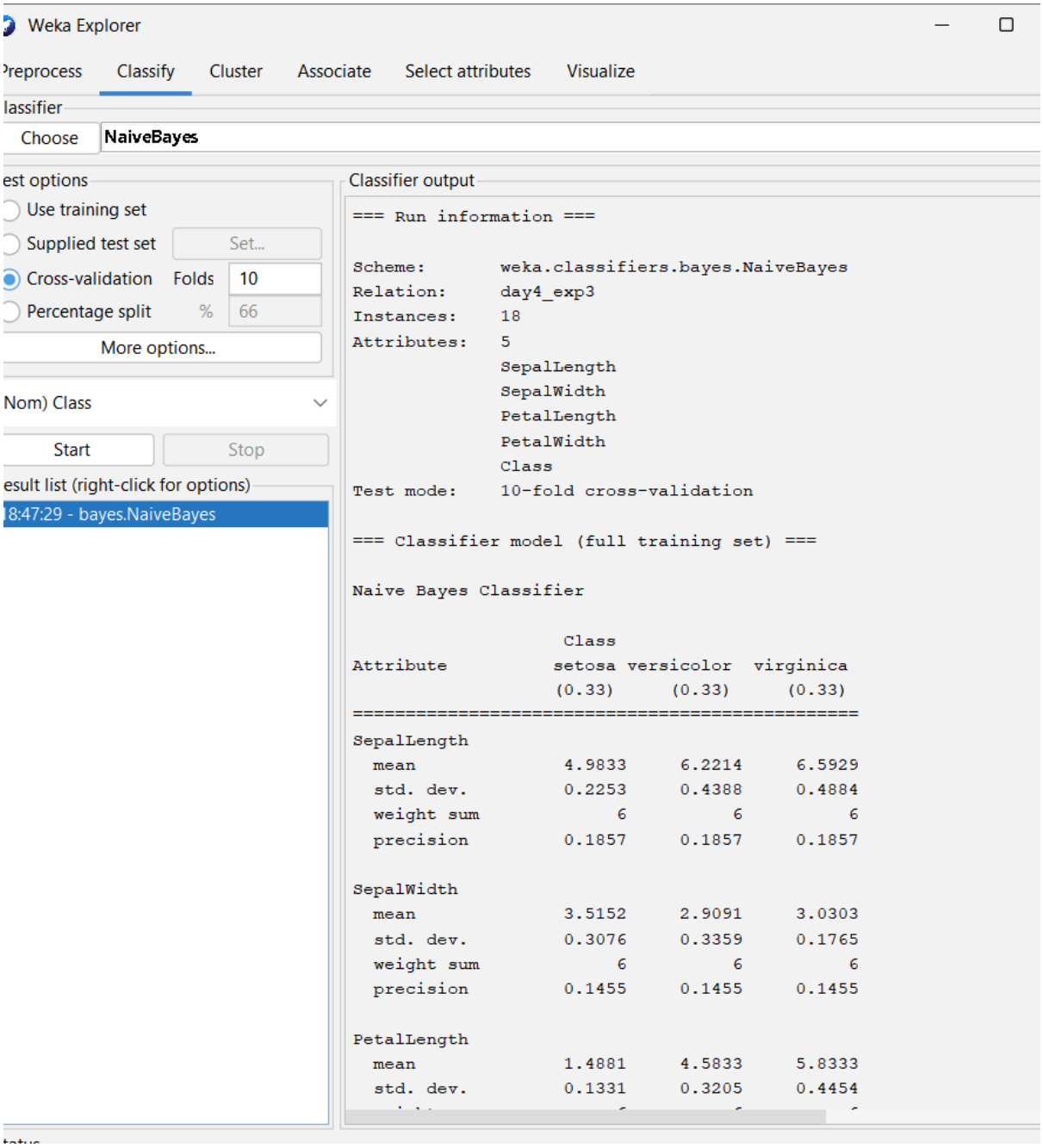
Aim

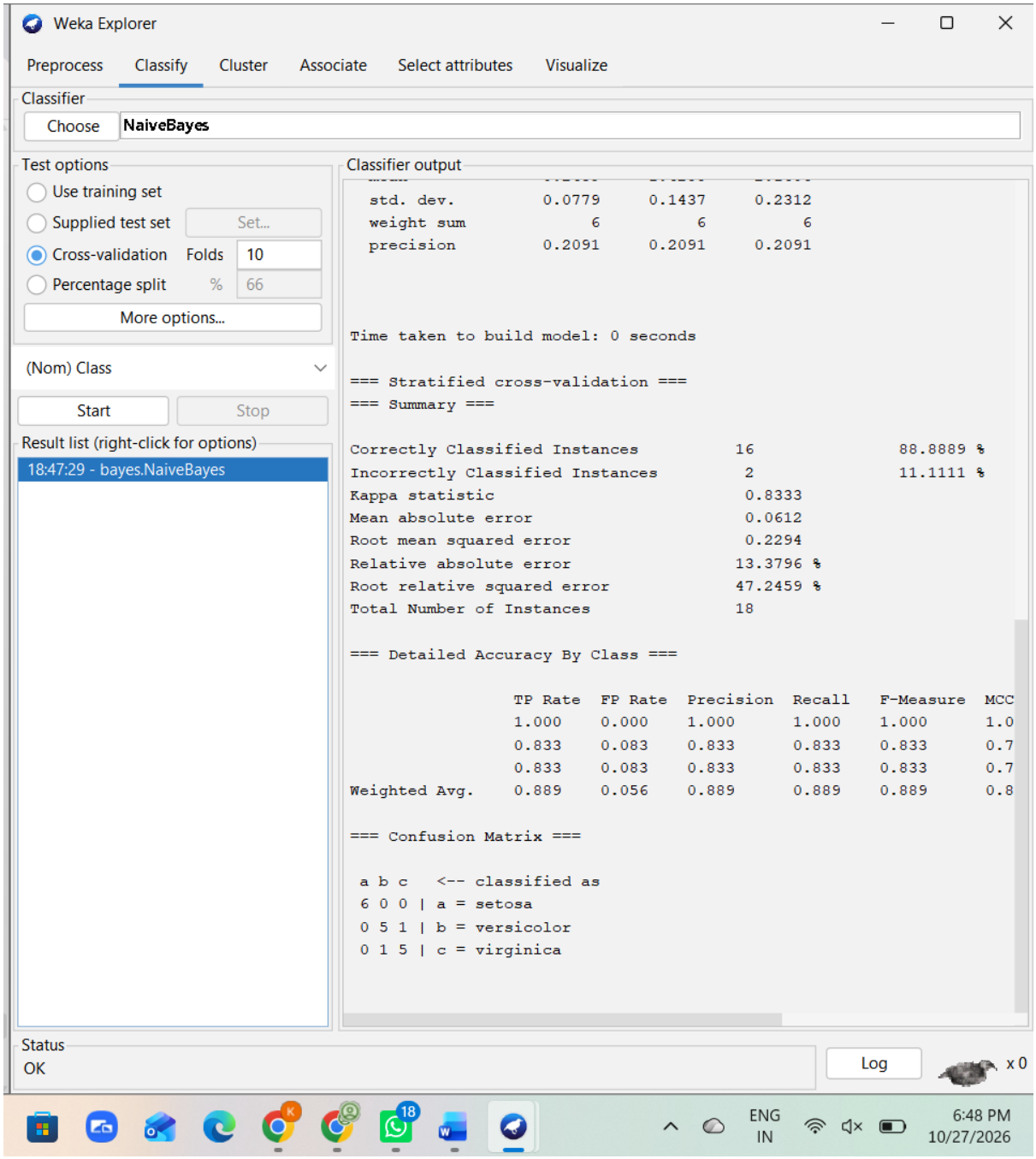
To perform categorical data prediction using the Naïve Bayes classification algorithm in WEKA, compare its performance with the Support Vector Machine (SVM) algorithm, and visualize the results.

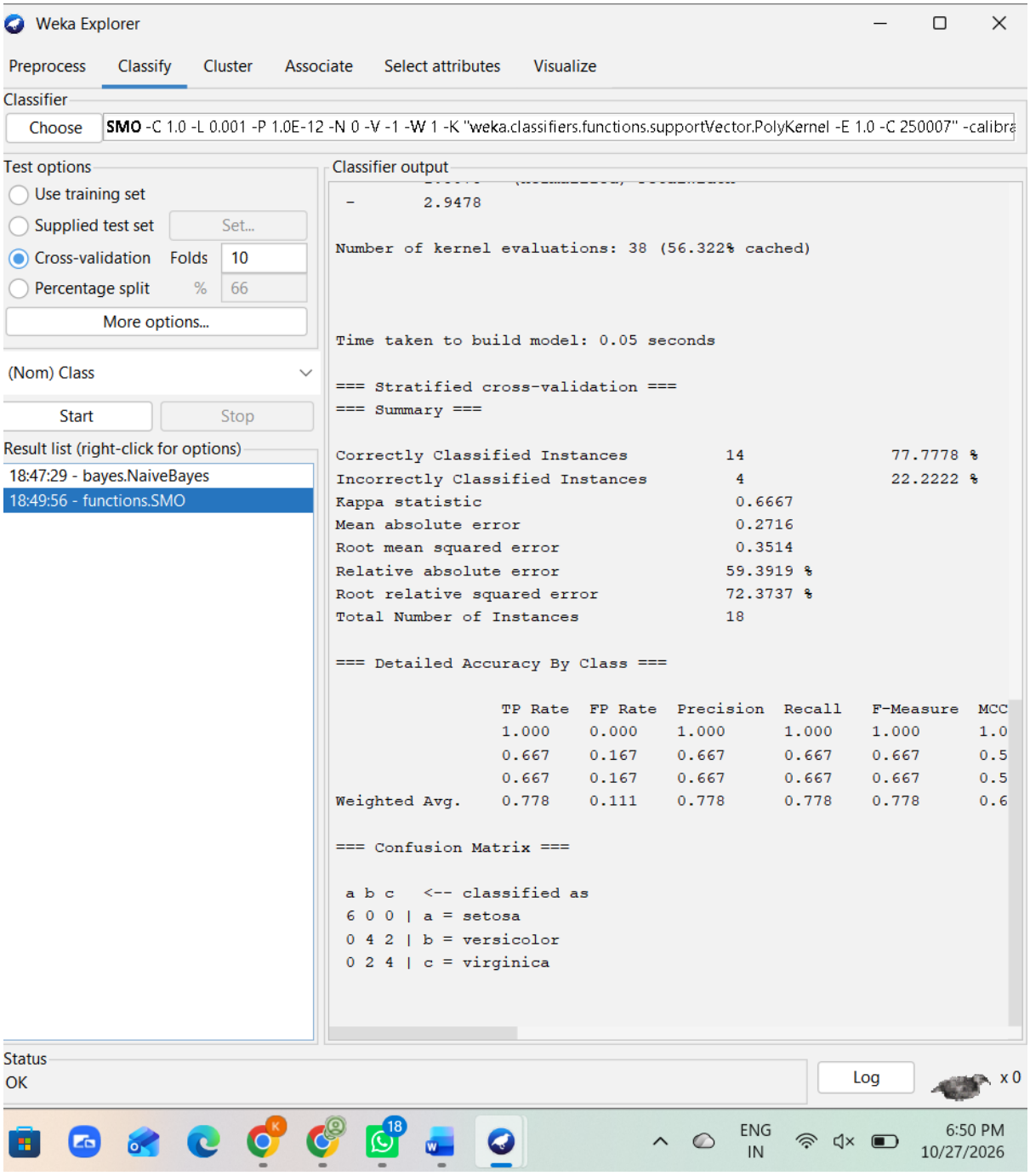
Procedure

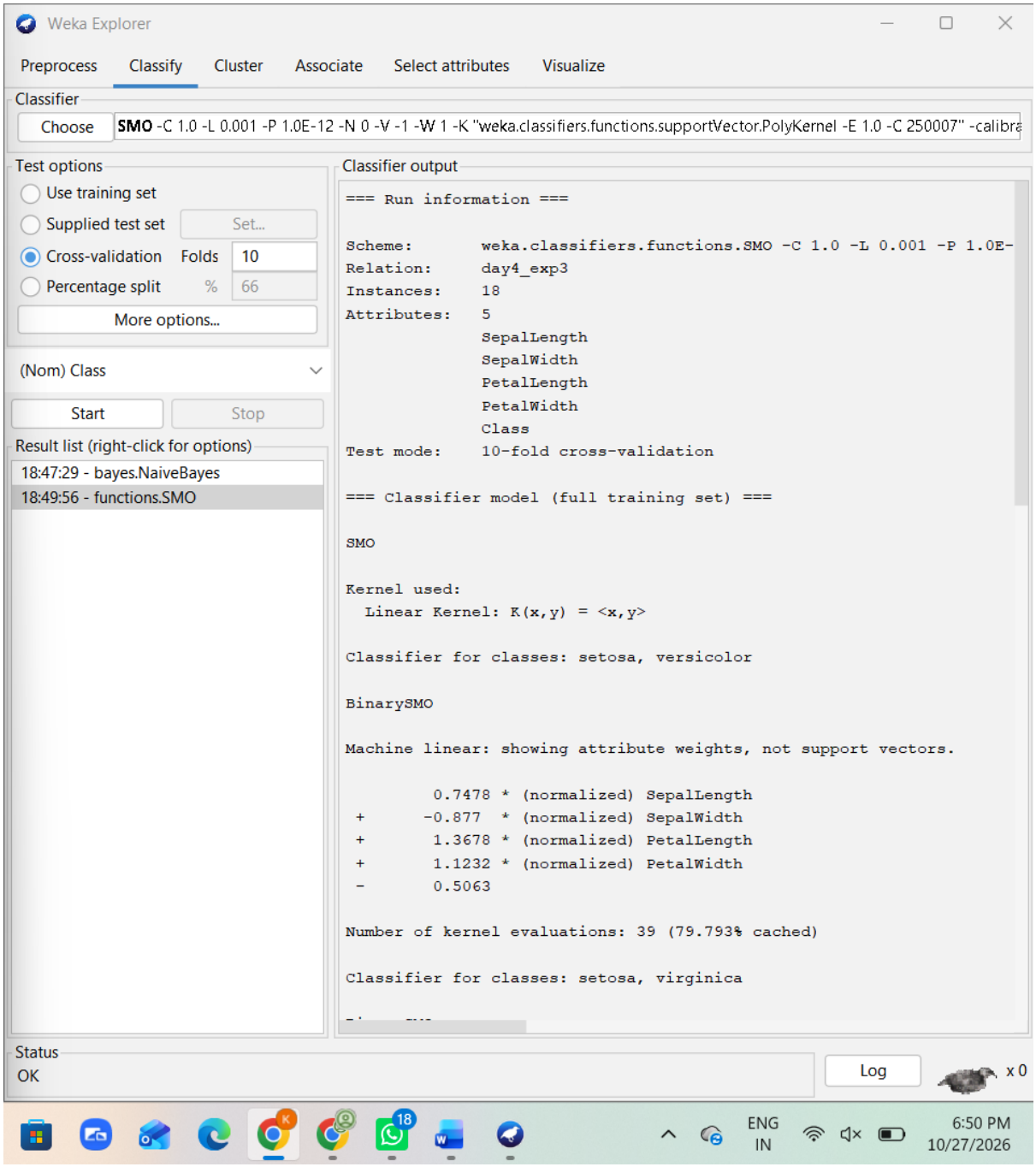
1. Open WEKA Explorer.
2. Click Preprocess → Open file and load day4_exp3.csv from your extracted datasets folder.
3. Check that the dataset contains a categorical target/class attribute.
4. Click the Classify tab.
5. Under Classifier, click Choose → bayes → NaiveBayes.
6. Under Test options, select Cross-validation and set the number of folds to 10.
7. Select the correct target attribute from the class dropdown.
8. Click Start and record the accuracy, confusion matrix, precision, recall and F-measure.
9. Select Choose → functions → SMO to run the SVM classifier.
10. Keep the same dataset, class attribute and cross-validation settings, then click Start.
11. Record the SVM results and compare them with the Naïve Bayes results.
12. Create a bar graph comparing the accuracy of both classifiers.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: day4_exp3
Instances: 18
Attributes: 5
Sum of weights: 18
Attributes
All
None
Invert
Pattern
No.
Name
1 SepalLength
2 SepalWidth
3 PetalLength
4 PetalWidth
5 Class
Remove
Selected attribute
Name: SepalLength
Missing: 0 (0%)
Distinct: 15
Type: Numeric
Unique: 12 (67%)
Statistic
Value
Minimum 4.7
Maximum 7.3
Mean 5.956
StdDev 0.814
Class: Class (Nom)
Visualize All
9
[4.7, 6]
Status
OK
Log
x 0
30°C
Mostly clear
Search
ENG
IN
6:45 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
lassifier
Choose
NaiveBayes
est options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
Nom) Class
Start
Stop
esult list (right-click for options)
18:47:29 - bayes.NaiveBayes
Classifier output
=== Run information ===
Scheme: weka.classifiers.bayes.NaiveBayes
Relation: day4_exp3
Instances: 18
Attributes: 5
SepalLength
SepalWidth
PetalLength
PetalWidth
Class
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
Naive Bayes Classifier
Class
Attribute setosa versicolor virginica
(0.33) (0.33) (0.33)
SepalLength
mean 4.9833 6.2214 6.5929
std. dev. 0.2253 0.4388 0.4884
weight sum 6 6 6
precision 0.1857 0.1857 0.1857
SepalWidth
mean 3.5152 2.9091 3.0303
std. dev. 0.3076 0.3359 0.1765
weight sum 6 6 6
precision 0.1455 0.1455 0.1455
PetalLength
mean 1.4881 4.5833 5.8333
std. dev. 0.1331 0.3205 0.4454


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
NaiveBayes
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Class
Start
Stop
Result list (right-click for options)
18:47:29 - bayes.NaiveBayes
Classifier output
std. dev. 0.0779 0.1437 0.2312
weight sum 6 6 6
precision 0.2091 0.2091 0.2091
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 16 88.8889 %
Incorrectly Classified Instances 2 11.1111 %
Kappa statistic 0.8333
Mean absolute error 0.0612
Root mean squared error 0.2294
Relative absolute error 13.3796 %
Root relative squared error 47.2459 %
Total Number of Instances 18
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC
1.000 0.000 1.000 1.000 1.000 1.0
0.833 0.083 0.833 0.833 0.833 0.7
0.833 0.083 0.833 0.833 0.833 0.7
Weighted Avg. 0.889 0.056 0.889 0.889 0.889 0.8
=== Confusion Matrix ===
a b c <-- classified as
6 0 0 | a = setosa
0 5 1 | b = versicolor
0 1 5 | c = virginica
Status
OK
Log
x 0
ENG
IN
6:48 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibra
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Class
Start
Stop
Result list (right-click for options)
18:47:29 - bayes.NaiveBayes
18:49:56 - functions.SMO
Classifier output
- 2.9478
Number of kernel evaluations: 38 (56.322% cached)
Time taken to build model: 0.05 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 14 77.7778 %
Incorrectly Classified Instances 4 22.2222 %
Kappa statistic 0.6667
Mean absolute error 0.2716
Root mean squared error 0.3514
Relative absolute error 59.3919 %
Root relative squared error 72.3737 %
Total Number of Instances 18
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC
1.000 0.000 1.000 1.000 1.000 1.0
0.667 0.167 0.667 0.667 0.667 0.5
0.667 0.167 0.667 0.667 0.667 0.5
Weighted Avg. 0.778 0.111 0.778 0.778 0.778 0.6
=== Confusion Matrix ===
a b c <-- classified as
6 0 0 | a = setosa
0 4 2 | b = versicolor
0 2 4 | c = virginica
Status
OK
Log
x 0
ENG
IN
6:50 PM
10/27/2026



Result

Naïve Bayes and SVM classification are performed using WEKA. Their performance is compared using accuracy, precision, recall, F-measure and confusion matrices. The results are used to identify which classifier performs better on the selected dataset.

4.The following list of persons with vegetarian or not details given in the table. How will you find out how many of them are vegetarian and how many of them are non-vegetarian? Which type of the person total count is greater value?

Person	Gopu	Babu	Baby	Gopal	Krishna	Jai	Dev	Malini	Hema	Anu
Vegetarian	yes	yes	yes	no	yes	no	no	yes	yes	yes

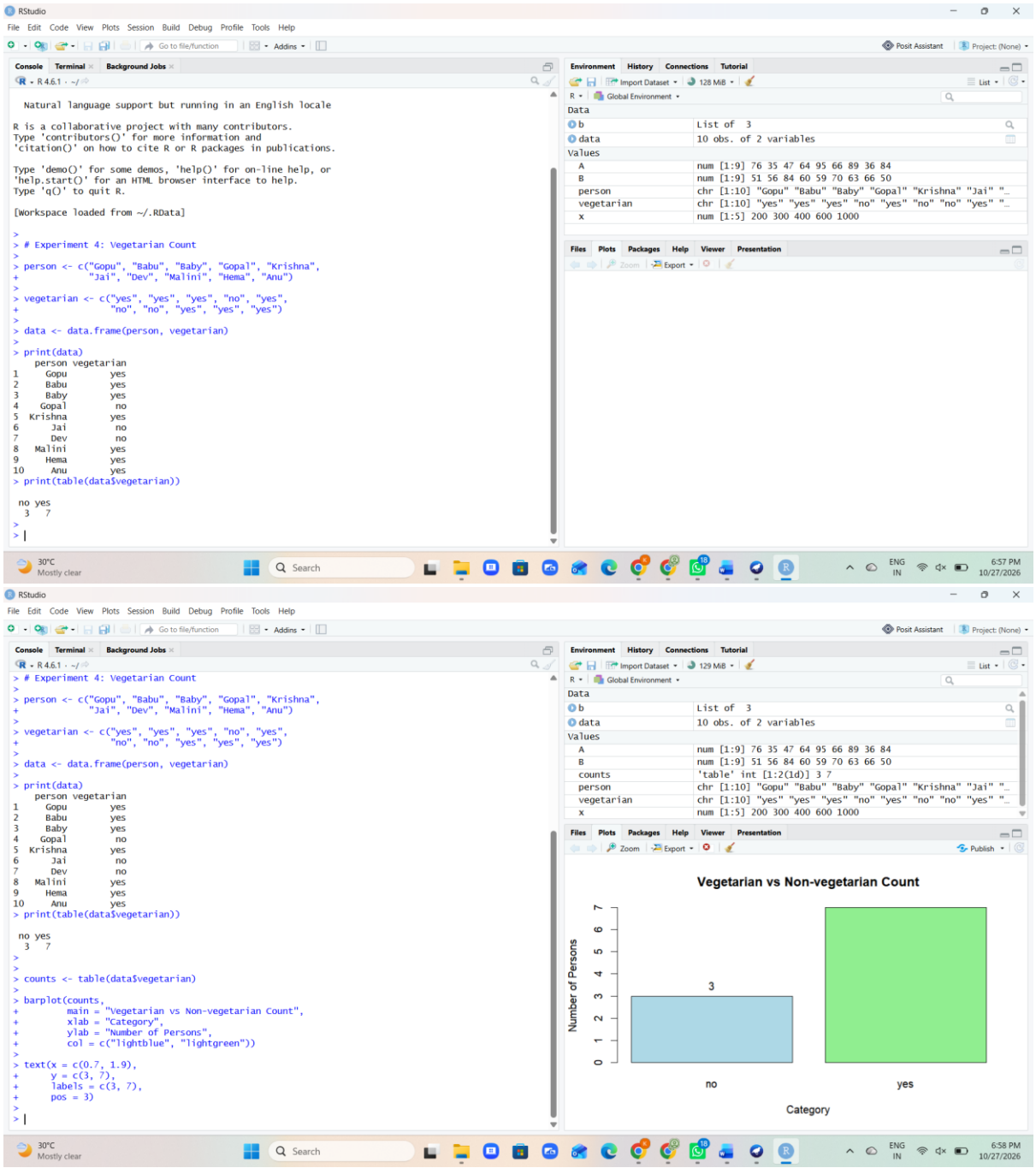
Aim

To count the number of vegetarian and non-vegetarian persons from the given dataset and identify which category has the higher count using R programming.

x	4	1	5	7	10	2	50	25	90	36
y	12	5	13	19	31	7	153	72	275	110

Procedure

1. Open RStudio.
2. Create a dataset containing the names of 10 persons and their vegetarian status.
3. Store the values as yes for vegetarian and no for non-vegetarian.
4. Create a data frame using the data.frame() function.
5. Use the table() function to count the number of vegetarians and non-vegetarians.
6. Use the barplot() function to represent the counts graphically.
7. Observe the output and compare the two categories.



Result

The experiment was successfully performed using RStudio. Out of 10 persons, 7 are vegetarians and 3 are non-vegetarians. Therefore, the number of vegetarians is greater than the number of non-vegetarians.

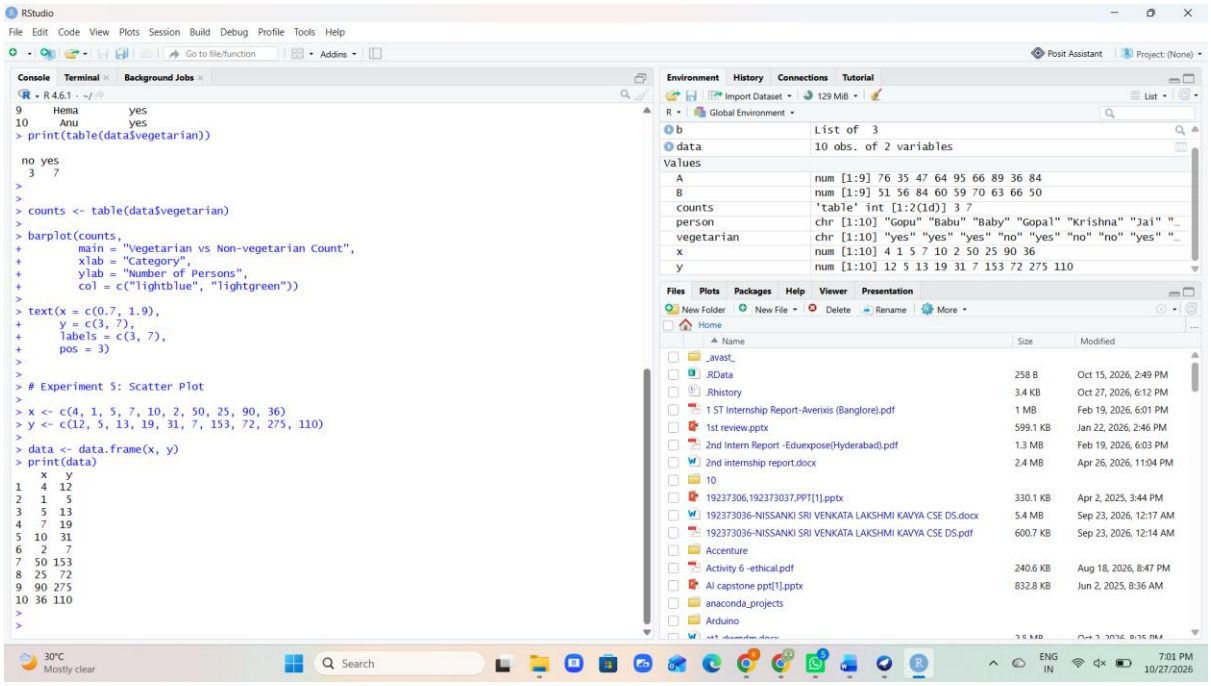
5.The following table would be plotted as (x,y) points, with the first column being the x values as number of mobile phones sold and the second column being the y values as money. To use the scatter plot for how many mobile phones sold.

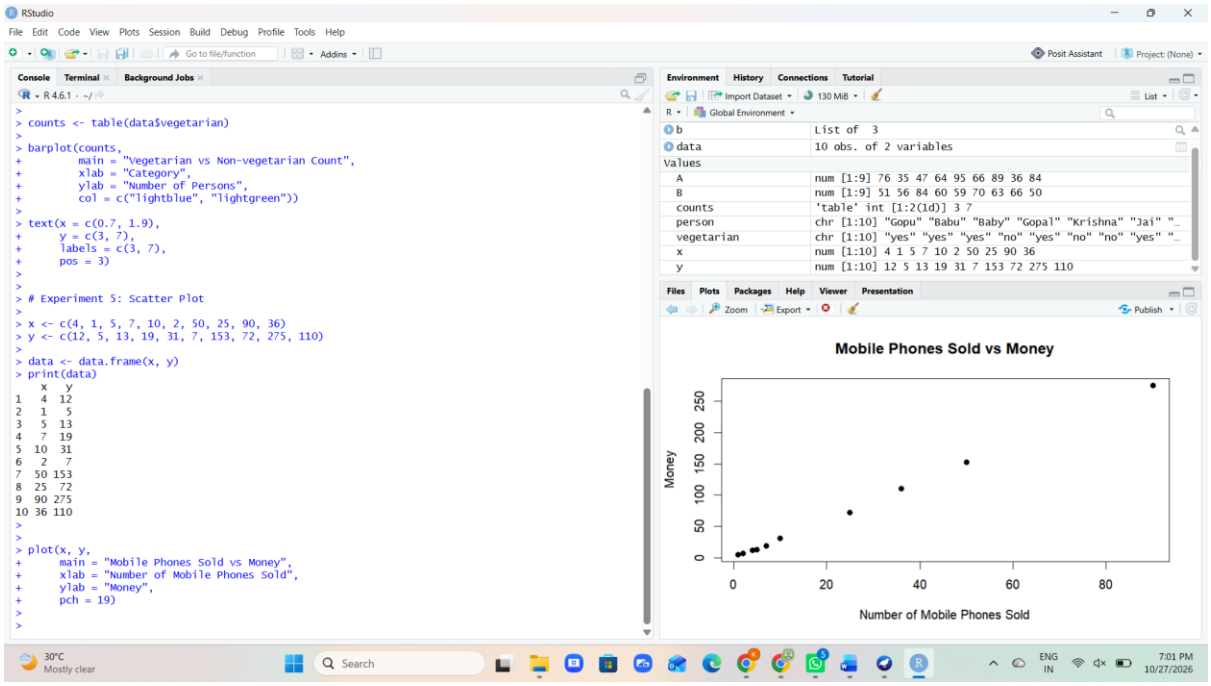
Aim

To plot a scatter graph using the given data to show the relationship between the number of mobile phones sold and the money earned.

Procedure

1. Open RStudio.
2. Enter the given X and Y values in two vectors.
3. Create the X values using c(4, 1, 5, 7, 10, 2, 50, 25, 90, 36).
4. Create the Y values using c(12, 5, 13, 19, 31, 7, 153, 72, 275, 110).
5. Use the plot() function to create the scatter plot.
6. Add a suitable title and label the X-axis as Number of Mobile Phones Sold and the Y-axis as Money.
7. View the graph in the Plots pane and save it.





x	4	1	5	7	10	2	50	25	90	36
y	12	5	13	19	31	7	153	72	275	110

Result

The scatter plot was successfully generated in RStudio using the given data. It represents the relationship between the number of mobile phones sold and the corresponding money earned.

6.Generate rules using FP growth algorithm using the given dataset which has the following transactions with items purchased: Consider the values as support=50% and confidence=75%.

Transaction ID	Items Purchased
1	Bread, Cheese, Egg, Juice
2	Bread, Cheese, Juice
3	Bread, Milk, Yogurt
4	Bread, Juice, Milk
5	Cheese, Juice, Milk

Aim

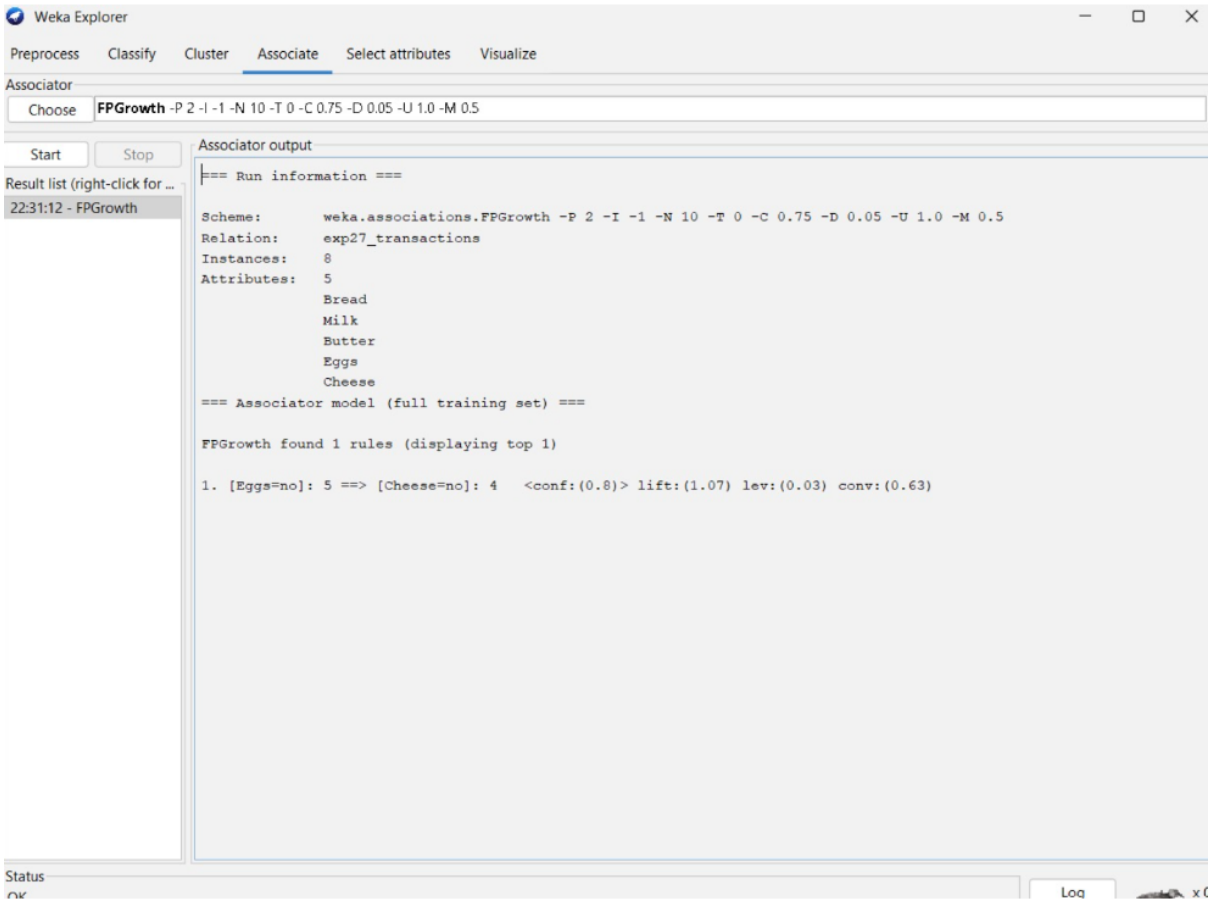
To generate frequent itemsets and association rules using the FP-Growth algorithm in WEKA with a minimum support of 50% and minimum confidence of 75%.

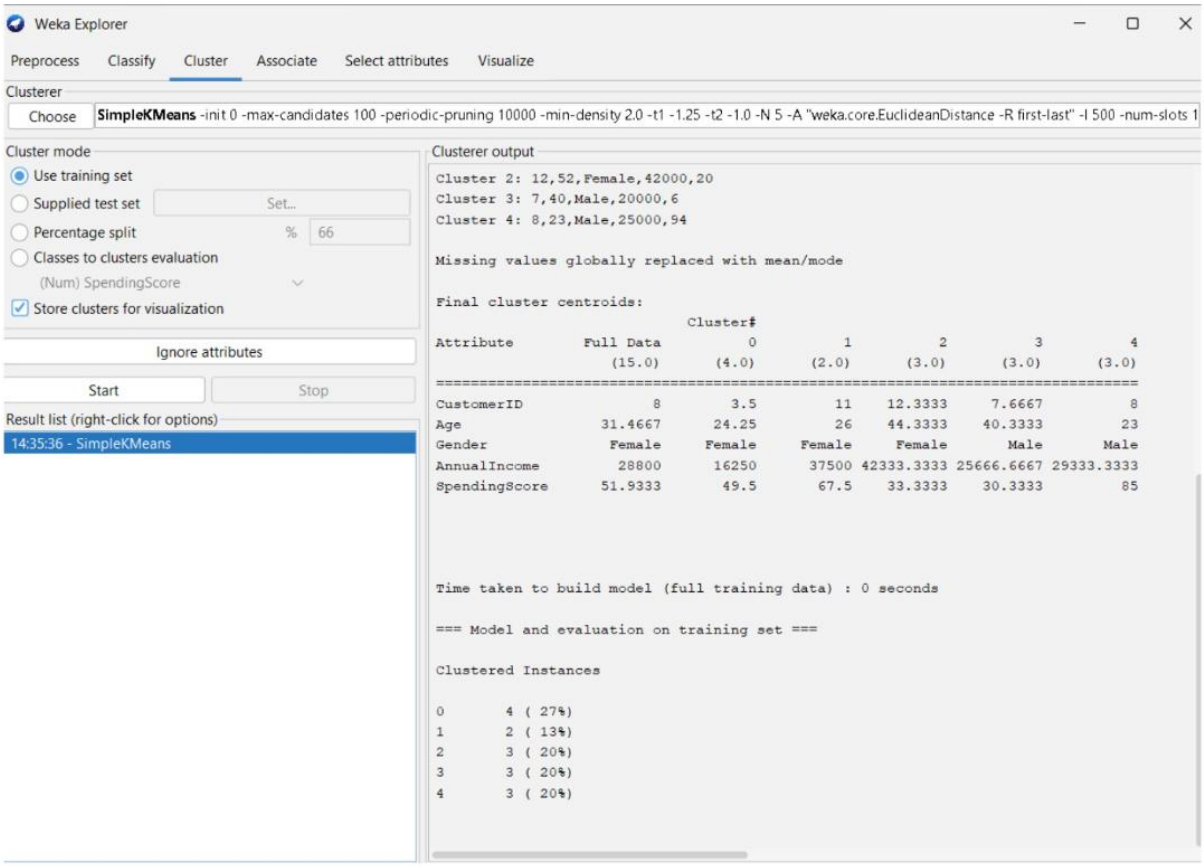
Procedure

1. Open WEKA Explorer.
2. Click the Preprocess tab.

3. Click Open file and load the dataset provided for Experiment 6.
4. Check that the dataset contains transaction items in the appropriate nominal format.
5. Click the Associate tab.
6. Click Choose and select FPGrowth, if available.
7. Click the algorithm name to open its settings.
8. Set the minimum support to 0.5 (50%) and minimum confidence to 0.75 (75%), using the options available in your WEKA version.
9. Click OK, then click Start.
10. Observe the output and record the frequent itemsets and association rules.
11. Take a screenshot of the WEKA output.

Figure 6: Association Rules Generated Using FP-Growth in WEKA.





Result

The FP-Growth algorithm was used to identify frequent itemsets and generate association rules from the transaction dataset using the specified support and confidence thresholds.

7.Prediction of Diabetes Data using Decision tree classifier in WEKA. Compare it with Support Vector Machine classifier. Show the result accuracy and F1 measure calculation .Plot the graph and explain the summary of results.

Aim

To predict diabetes using a Decision Tree classifier in WEKA, compare its performance with the Support Vector Machine (SVM), calculate accuracy and F-measure, and plot a comparison graph.

Procedure

Step 1: Load the dataset

1. Open WEKA Explorer.
2. Click Preprocess → Open file.

3. Select your Experiment 7 diabetes dataset (day4_exp7.arff or the supplied equivalent).
4. Check that the dataset is loaded correctly and contains a diabetes class attribute.

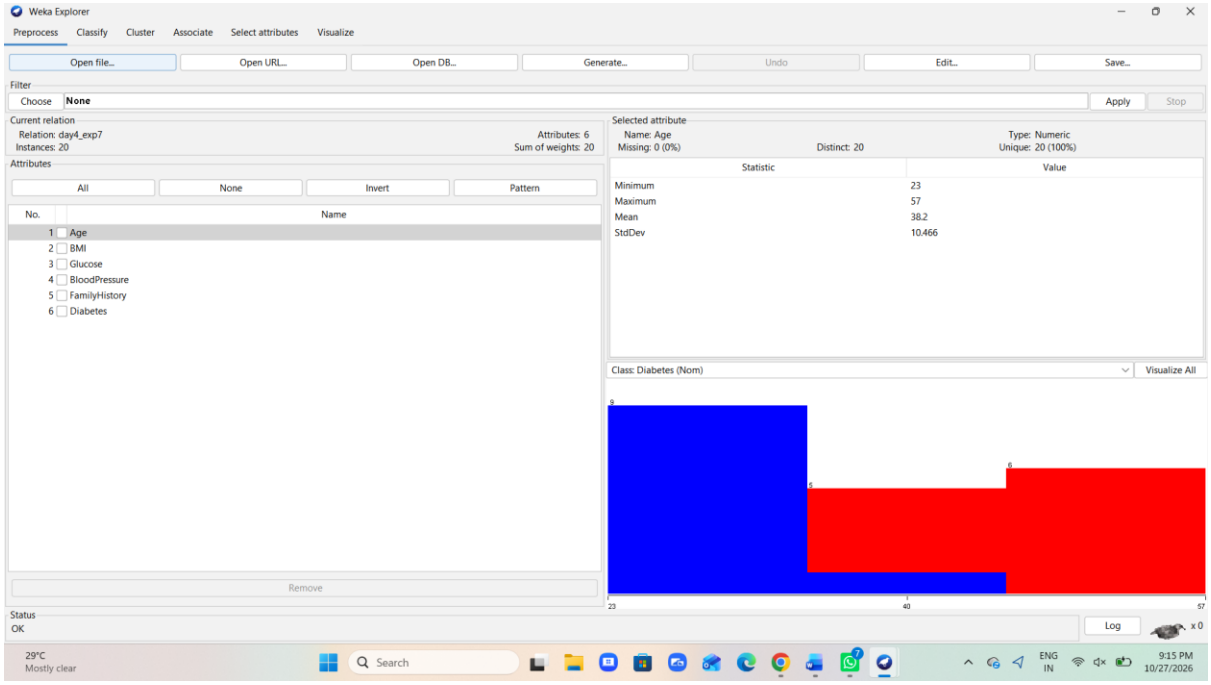
Step 2: Run the Decision Tree

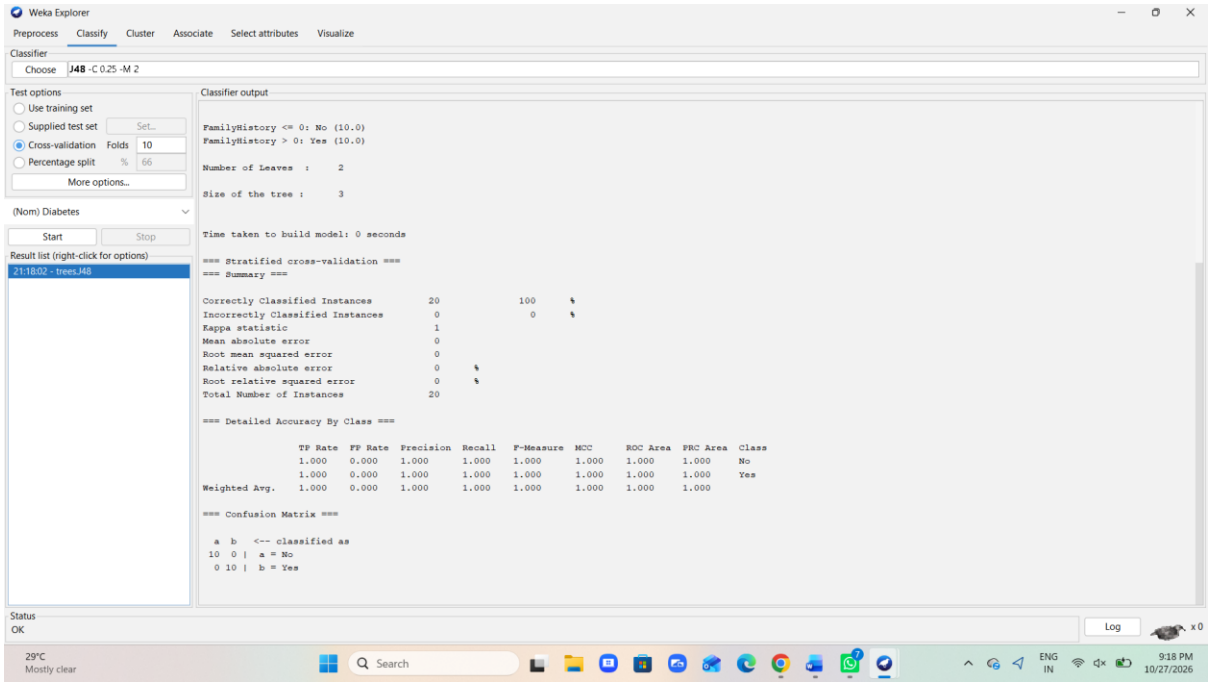
1. Click Classify.
2. Click Choose → trees → J48.
3. Under Test options, select Cross-validation and set folds to 10.
4. Select the diabetes outcome column as the Class attribute.
5. Click Start.
6. Record accuracy, precision, recall, F-measure and the confusion matrix.

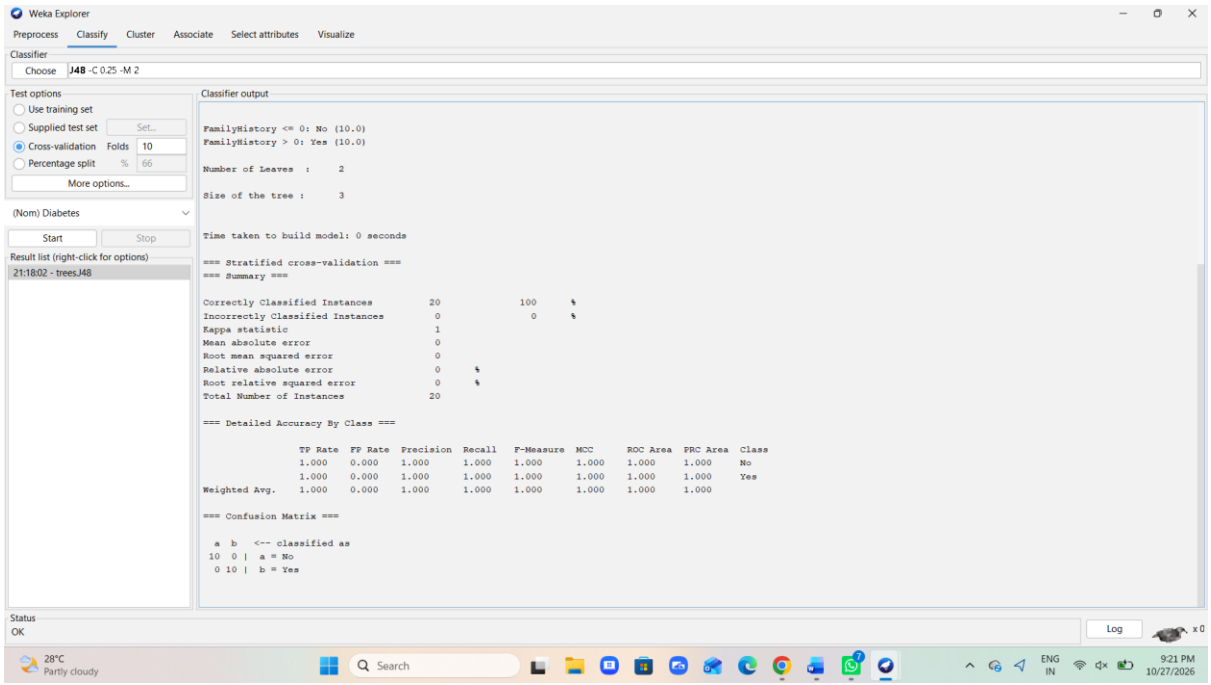
Step 3: Run SVM

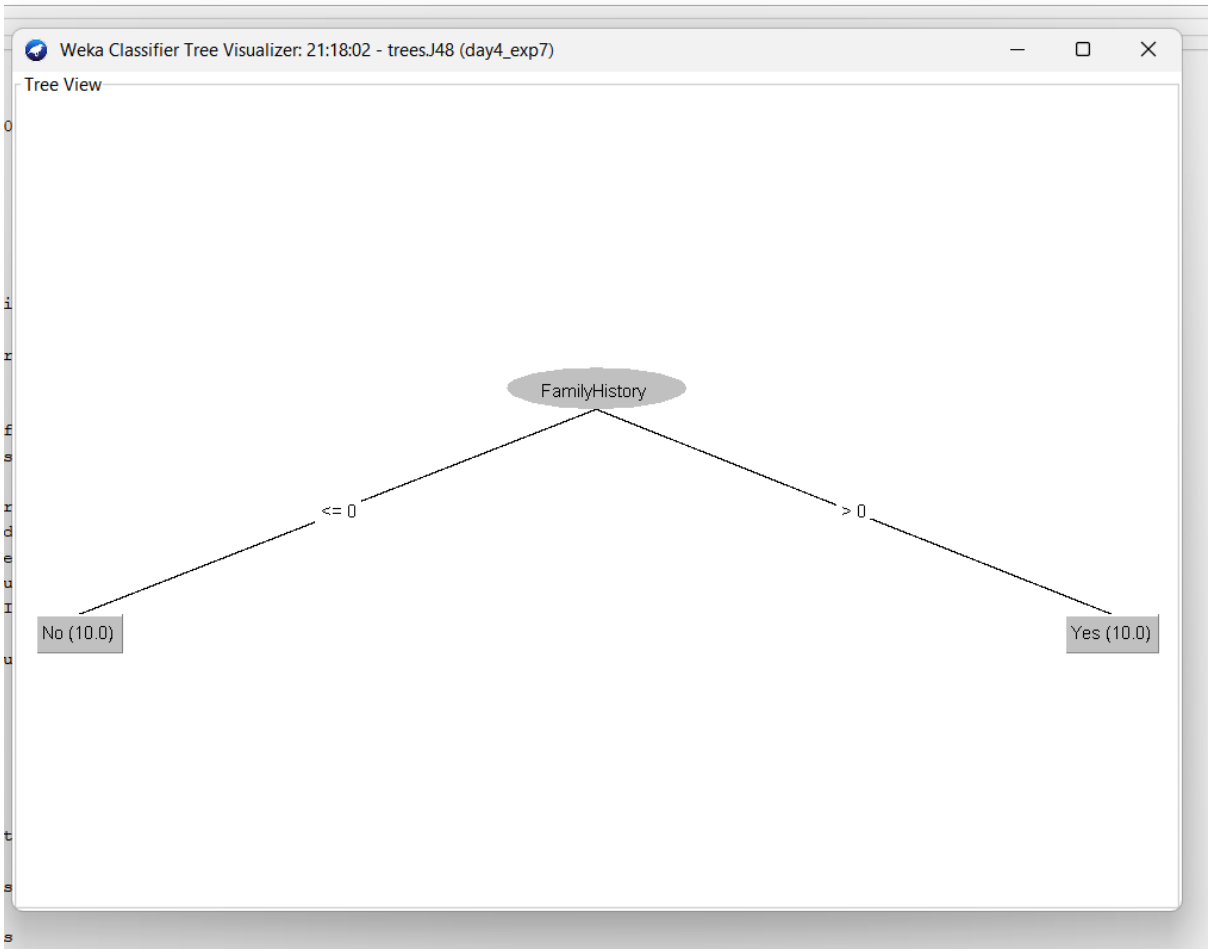
1. Click Choose → functions → SMO.
2. Keep the same dataset, class attribute and 10-fold cross-validation.
3. Click Start.
4. Record the same evaluation metrics.

Create a bar graph comparing the accuracy of both algorithms and save it.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Diabetes
Start
Stop
Result list (right-click for options)
21:18:02 - trees.J48
Classifier output
FamilyHistory <= 0: No (10.0)
FamilyHistory > 0: Yes (10.0)
Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 20 100 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
Total Number of Instances 20
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 No
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Yes
Weighted Avg. 1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000
=== Confusion Matrix ===
a b <-- classified as
10 0 | a = No
0 10 | b = Yes
Status
OK
Log
x 0
29°C
Mostly clear
Search
ENG
IN
9:18 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Diabetes
Start
Stop
Result list (right-click for options)
21:18:02 - trees.J48
Classifier output
FamilyHistory <= 0: No (10.0)
FamilyHistory > 0: Yes (10.0)
Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 20 100 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
Total Number of Instances 20
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 No
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Yes
Weighted Avg. 1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000
=== Confusion Matrix ===
a b <-- classified as
10 0 | a = No
0 10 | b = Yes
Status
OK
Log
x 0
28°C
Partly cloudy
Search
ENG
IN
9:21 PM
10/27/2026



Result

The Decision Tree and SVM classifiers are evaluated using the diabetes dataset in WEKA. Their accuracy and F-measure are compared to determine which classifier performs better on the selected dataset.

8.Implement of the R script using marks scored by a student in his model exam has been sorted as follows: 55, 60, 71, 63, 55, 65, 50, 55,58,59,61,63,65,67,71,72,75. Partition them into three bins by each of the following methods. Plot the data points using histogram.

(a) equal-frequency (equi-depth) partitioning
(b) equal-width partitioning
(c) clustering

Aim

To partition the given student marks into three bins using equal-frequency partitioning, equal-width partitioning, and clustering, and plot the data using histograms.

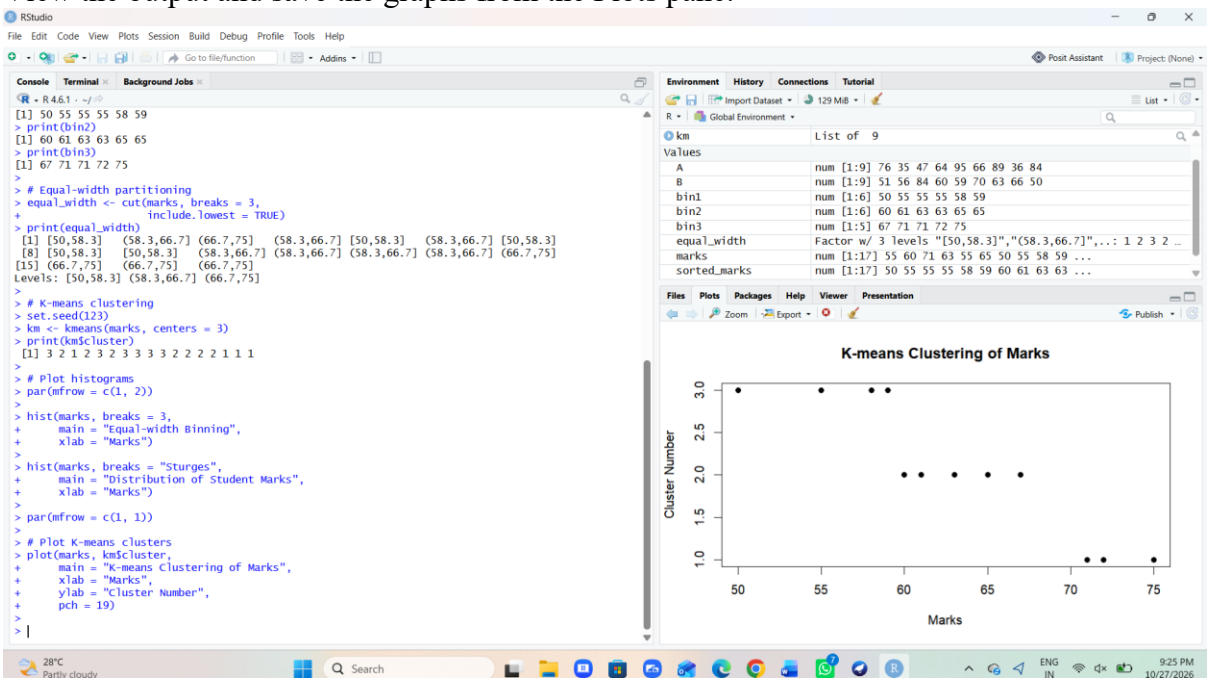
Given Data

Marks scored by the student:
55, 60, 71, 63, 55, 65, 50, 55, 58, 59, 61, 63, 65, 67, 71, 72, 75

Procedure

1. Open RStudio.
2. Enter the given student marks into a vector using the c() function.
3. Sort the marks in ascending order using sort().

4. Divide the sorted data into three bins using equal-frequency partitioning.
5. Divide the data into three bins of equal width using the cut() function.
6. Apply the K-Means clustering algorithm with three clusters.
7. Plot histograms to visualize the data distribution for the binning methods.
8. View the output and save the graphs from the Plots pane.



Result

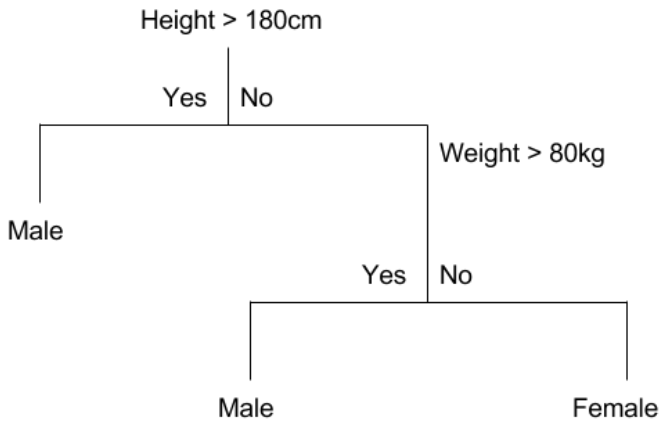
The student marks were divided into three bins using equal-frequency partitioning, equal-width partitioning, and K-Means clustering. The data distribution and clusters were visualized using graphs in RStudio.

9.Consider this Decision tree :

a)create the data set for the below tree using ARFF format and calculate accuracy and decision for the same

b) Using this decision tree generate the rules based on rule based induction.

c) Compare both the algorithms and plot the confusion matrix.



Aim

To create a dataset in ARFF format based on the given decision tree, calculate classification accuracy, generate IF–THEN rules using rule-based induction, and compare the algorithms using a confusion matrix.

Given Decision Tree

The decision tree in your lab document uses these conditions:

- Height greater than 180 cm → Male.
- Otherwise, check whether Weight is greater than 80 kg.
- If Weight is greater than 80 kg → Male.
- Otherwise → Female.

Procedure

Step 1: Create the dataset

1. Open Notepad.
2. Enter the following ARFF dataset.

Step 2: Load the dataset in WEKA

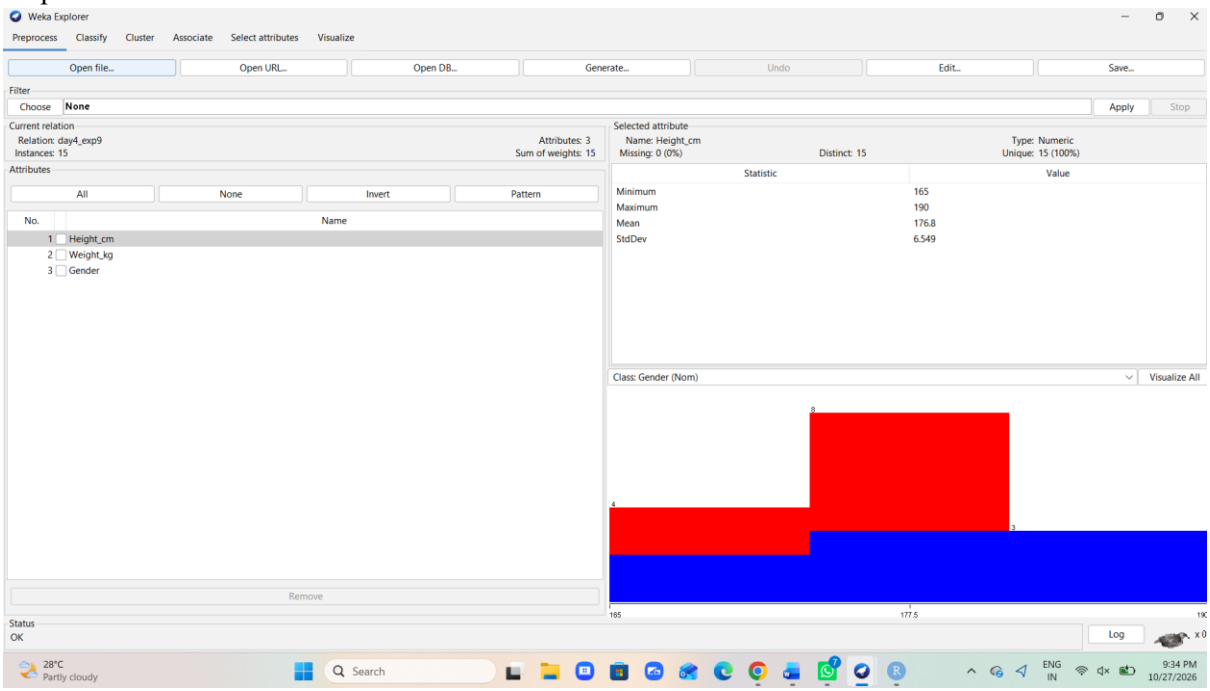
1. Open WEKA Explorer.
2. Click Preprocess → Open file.
3. Select day4_exp9.arff.
4. In the Classify tab, select trees → J48.
5. Choose Use training set and click Start.
6. Record the accuracy and confusion matrix.

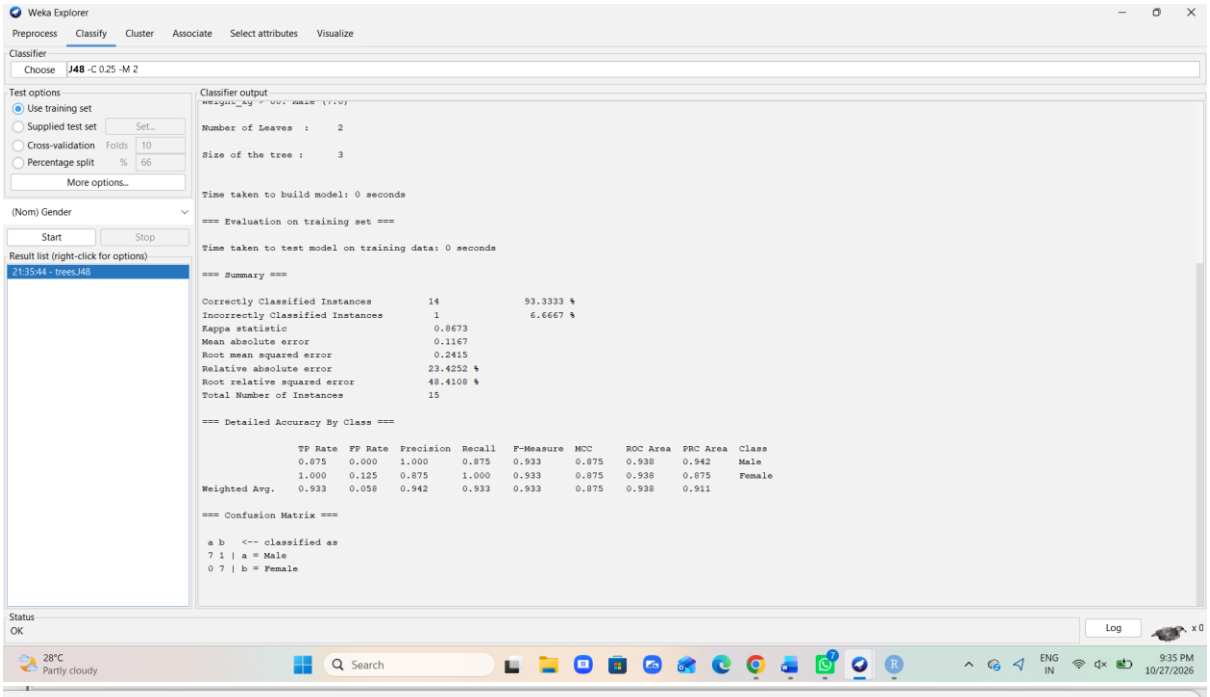
Step 3: Generate rules

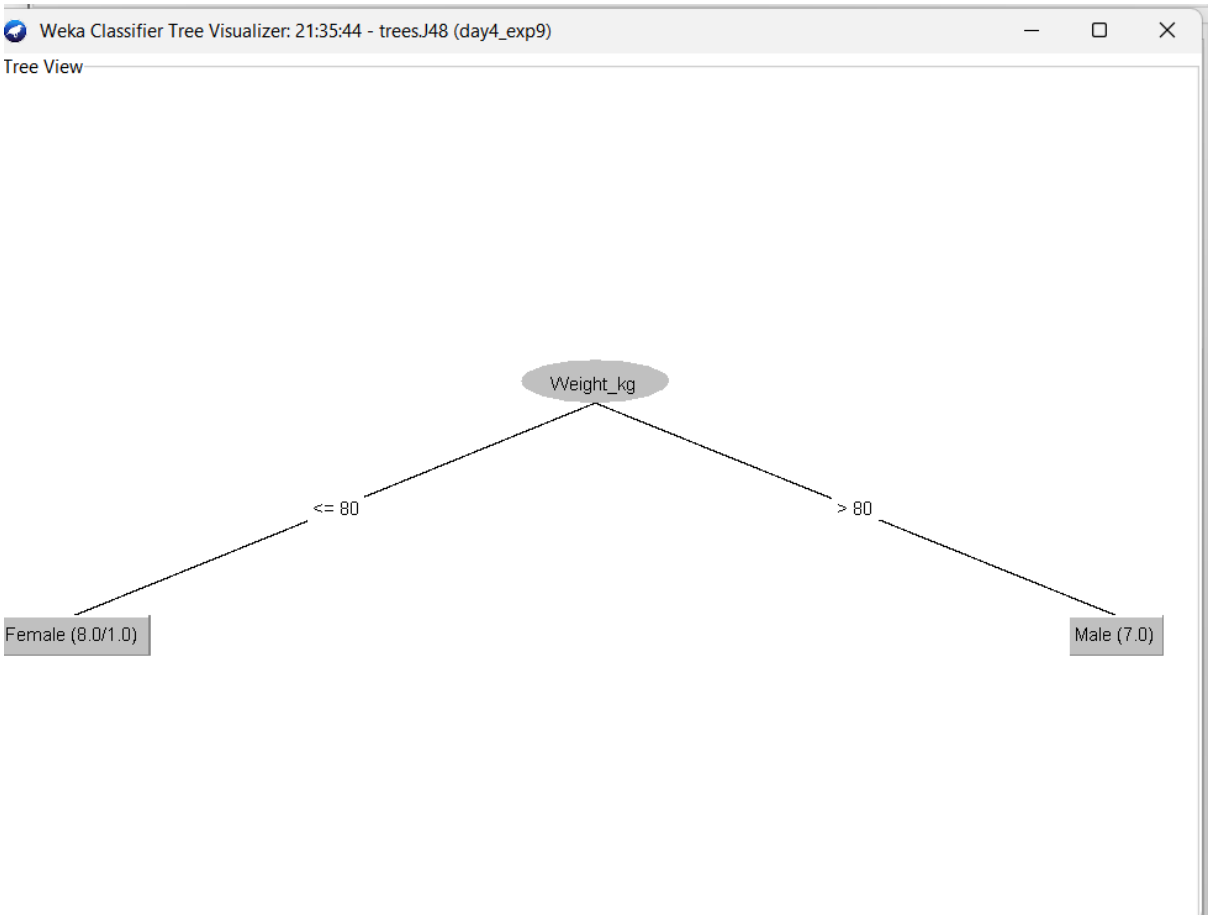
1. Click Choose → rules → JRip.
2. Keep the same dataset and class attribute.
3. Select Use training set and click Start.
4. Record the generated IF–THEN rules and classification results.

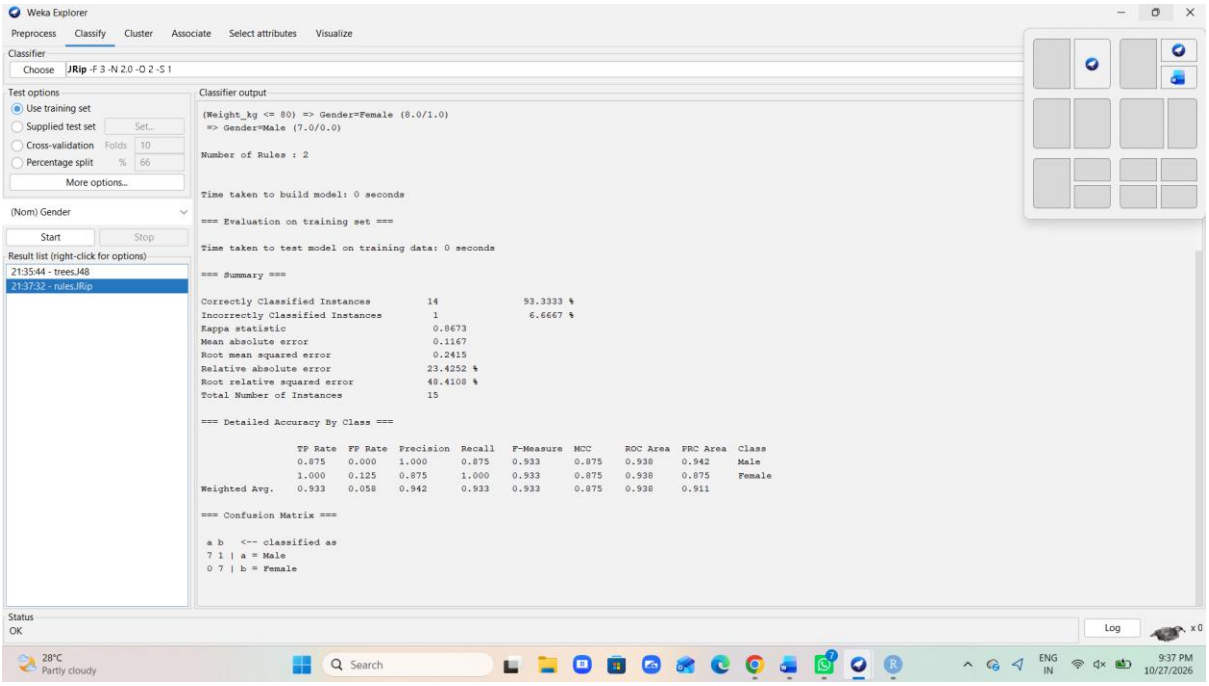
Step 4: Compare the algorithms

Compare J48 and JRip using their accuracy and confusion matrices. Capture screenshots of both outputs.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Gender
Start
Stop
Result list (right-click for options)
21:35:44 - trees.J48
Classifier output
Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0 seconds
=== Evaluation on training set ===
Time taken to test model on training data: 0 seconds
=== Summary ===
Correctly Classified Instances 14 93.3333 %
Incorrectly Classified Instances 1 6.6667 %
Kappa statistic 0.8673
Mean absolute error 0.1167
Root mean squared error 0.2415
Relative absolute error 23.4252 %
Root relative squared error 48.4108 %
Total Number of Instances 15
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.875 0.000 1.000 0.875 0.933 0.875 0.938 0.942 Male
1.000 0.125 0.875 1.000 0.933 0.875 0.938 0.875 Female
Weighted Avg. 0.933 0.058 0.942 0.933 0.933 0.875 0.938 0.911
=== Confusion Matrix ===
a b <-- classified as
7 1 | a = Male
0 7 | b = Female
Status
OK
Log
x 0
28°C
Partly cloudy
Search
ENG
IN
9:35 PM
10/27/2026


Weka Classifier Tree Visualizer: 21:35:44 - trees.J48 (day4_exp9)
Tree View
Weight_kg
<= 80
> 80
Female (8.0/1.0)
Male (7.0)



Result

The dataset was prepared in ARFF format, and the J48 decision tree and JRip rule-based classifier were used to examine gender

10.Create an ARFF file for the table below and implement for the Apriori Algorithm and FP growth algorithm and compare the rules generated by both the algorithms. Identify the unique rules generated by the above algorithms.

NOTE: Assume Min_sup=2 and confidence= 50%

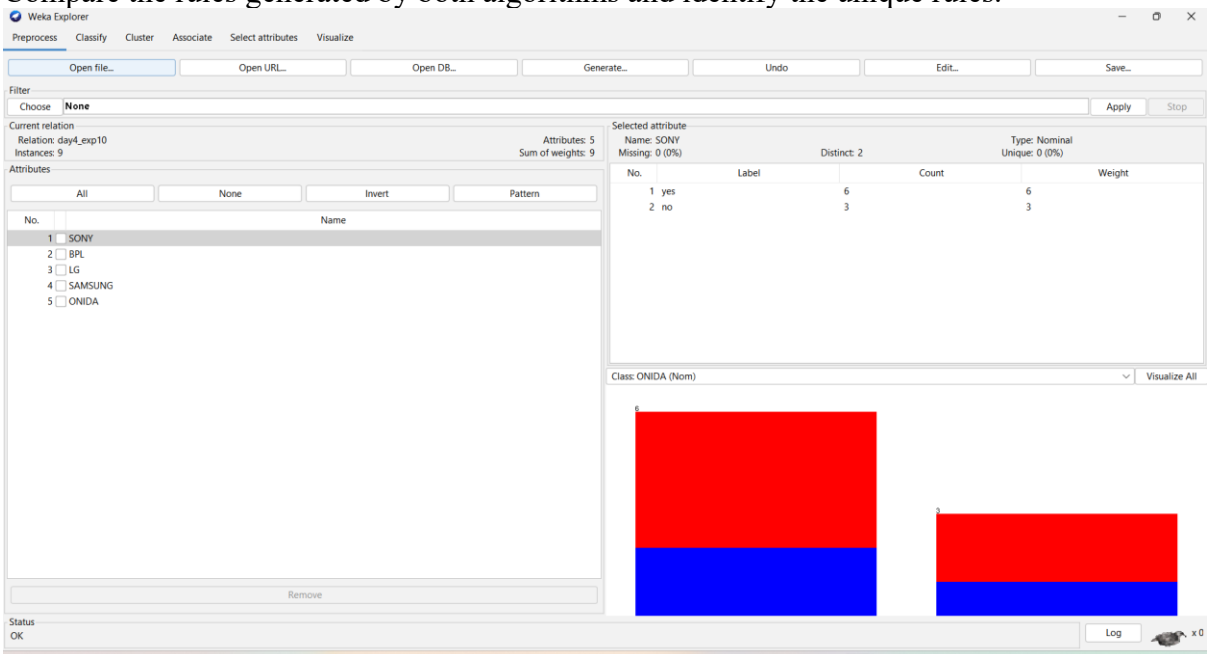
T.ID	ITEMS
T1	SONY, BPL, LG
T2	BPL, SAMSUNG
T3	BPL, ONIDA
T4	SONY, BPL, SAMSUNG
T5	SONY, ONIDA
T6	BPL, ONIDA
T7	SONY, ONIDA
T8	SONY, BPL, ONIDA, LG
T9	SONY, BPL, ONIDA

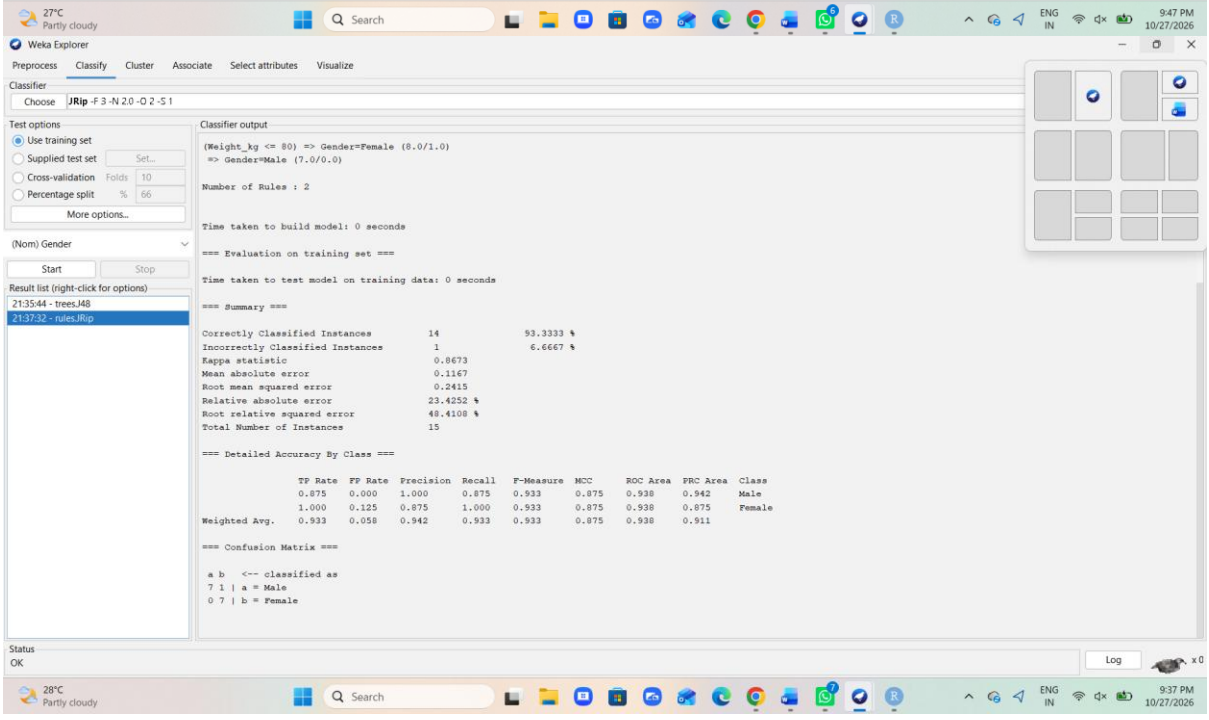
Aim

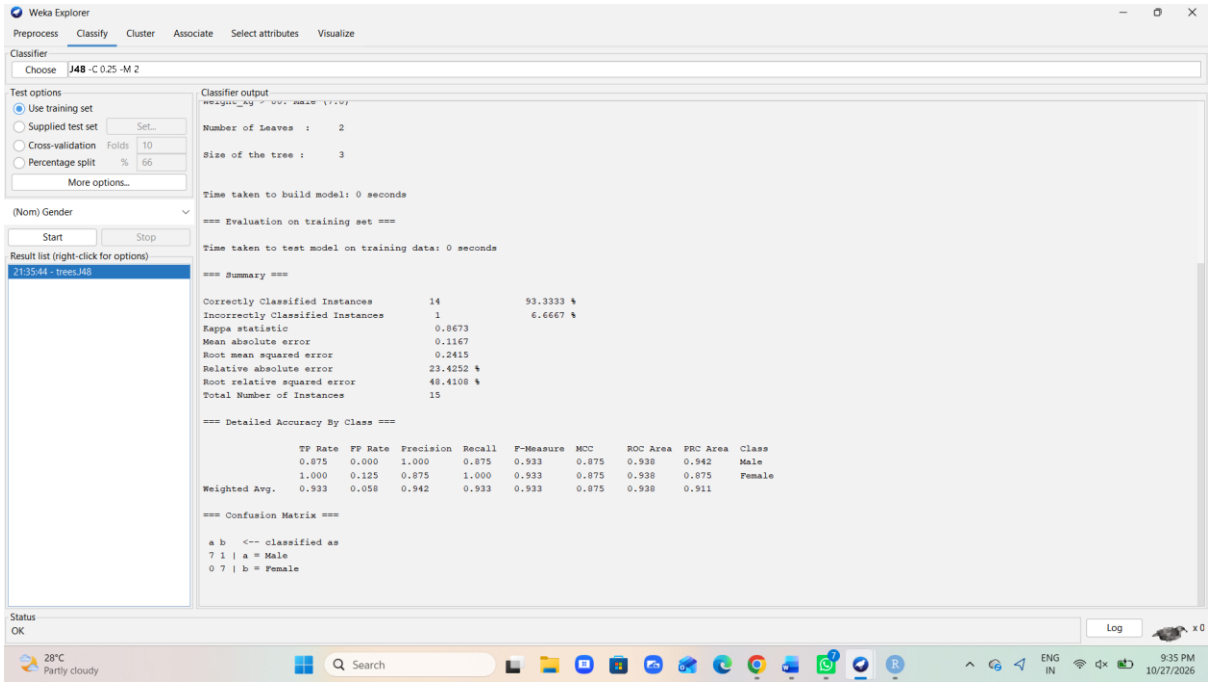
To implement the Apriori and FP-Growth algorithms using an ARFF dataset, compare the generated association rules, and identify the unique rules using minimum support of 2 and confidence of 50%.

Procedure

1. Open Notepad and prepare the transaction dataset containing the items purchased in each transaction.
2. Save the dataset in ARFF format as day4_exp10.arff.
3. Open WEKA Explorer and click the Preprocess tab.
4. Click Open file and load the saved ARFF dataset.
5. Go to the Associate tab and select Apriori.

6. Configure the minimum support count as 2 and confidence as 50%, using the available WEKA settings.
7. Click Start and record the frequent itemsets and association rules.
8. Select FPGrowth, if available, and run the algorithm on the same dataset with equivalent settings.
9. Compare the rules generated by both algorithms and identify the unique rules.
10. 

11. 



Result

The Apriori and FP-Growth algorithms are used to generate frequent itemsets and association rules from the transaction dataset. The generated rules are compared to identify common and unique association rules.

11,The given are the strike-rates scored by a batsman in season 1 in different tournaments.
100, 70, 60, 90, 90

(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization
(c) z-score normalization using the mean absolute deviation instead of standard deviation
(d) normalization by decimal scaling

Aim

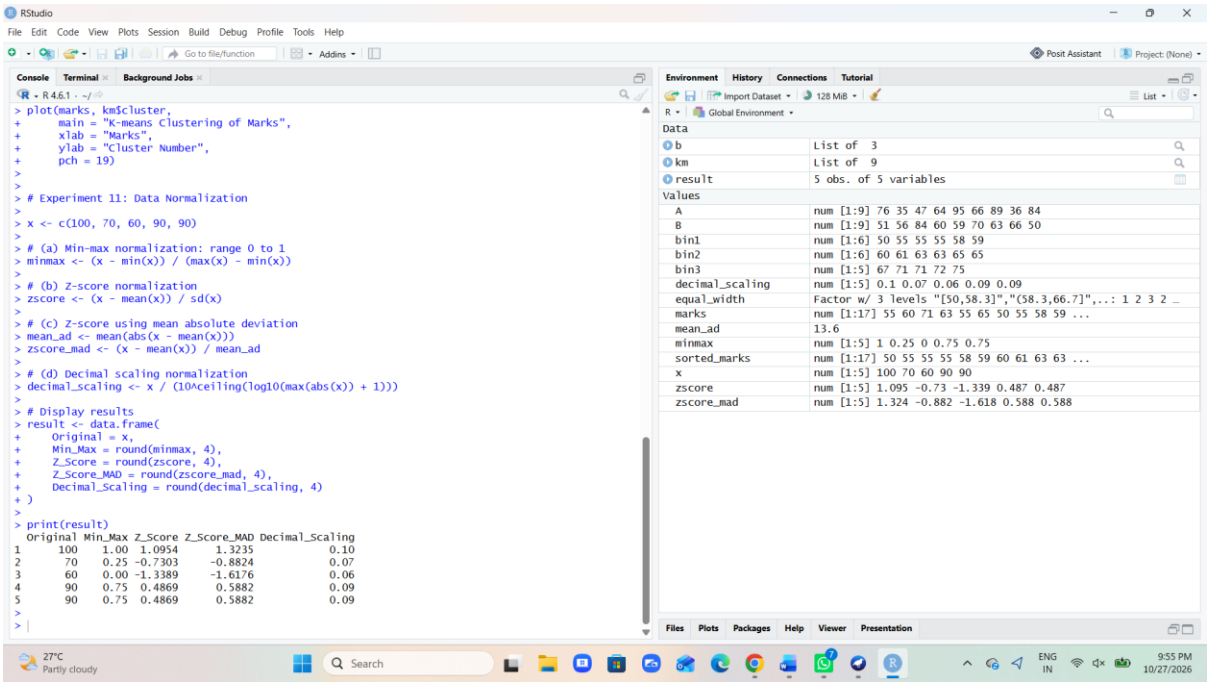
To perform min-max normalization, z-score normalization, z-score normalization using mean absolute deviation, and decimal-scaling normalization on the given batsman's strike rates using R.

Given data

100, 70, 60, 90, 90

Procedure

1. Open RStudio and create a new R script.
2. Enter the given strike-rate values into a vector.
3. Calculate min-max normalization by setting the minimum to 0 and maximum to 1.
4. Calculate z-score normalization using the mean and standard deviation.
5. Calculate z-score normalization using mean absolute deviation instead of standard deviation.
6. Perform decimal-scaling normalization by dividing each value by the appropriate power of 10.
7. Display the normalized values in a table.
8. Compare the results obtained using the four normalization methods.
9. Save the R script and take a screenshot of the output.



Result

The four normalization methods were implemented successfully in RStudio. The original strike-rate values were transformed using min-max normalization, z-score normalization, mean absolute deviation-based normalization, and decimal scaling.

12Suppose some car is tested for the AvgSpeed and TotalTime data for 9 randomly selected car with the following result

AvgSpeed (in kph)	**78**	**81**	**82**	**74**	**83**	**82**	**77**	**80**	**70**
TotalTime (in mins)	**39**	**37**	**36**	**42**	**35**	**36**	**40**	**38**	**46**

a) Calculate the standard deviation of AvgSpeed and TotalTime.

b) Calculate the Variance of AvgSpeed and TotalTime for the above dataset.

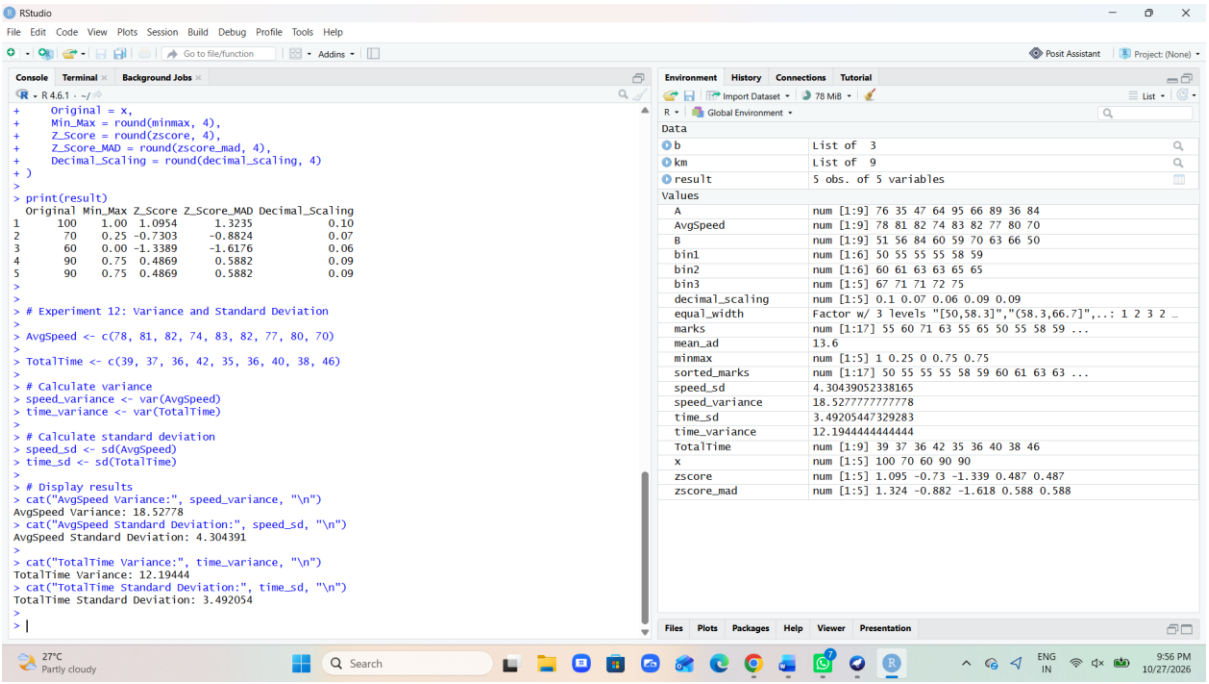
1. Aim

To calculate the variance and standard deviation of `AvgSpeed` and `TotalTime` using RStudio.

2. Procedure

c) Open RStudio.
d) Create a new R Script.
e) Enter the given data for `AvgSpeed` and `TotalTime`.
f) Calculate the variance using the `var()` function.

g) Calculate the standard deviation using the `sd()` function.
h) Run the program and observe the output.



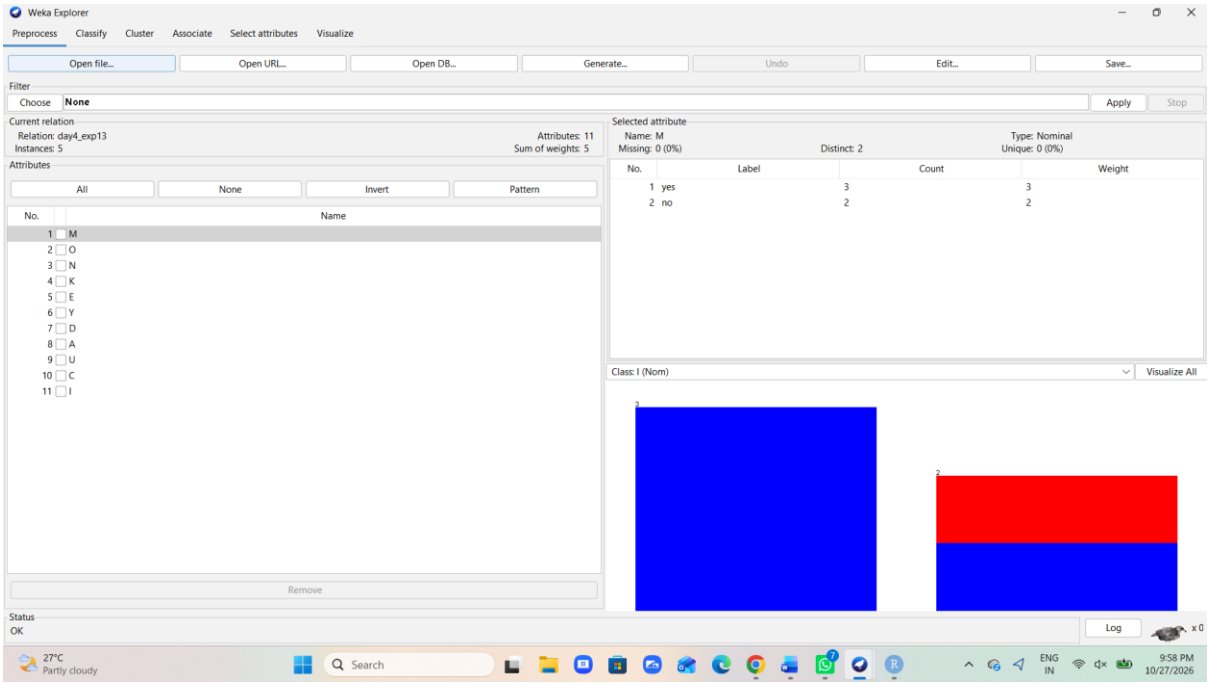
Result: The variance and standard deviation of AvgSpeed and TotalTime were calculated successfully using RStudio.

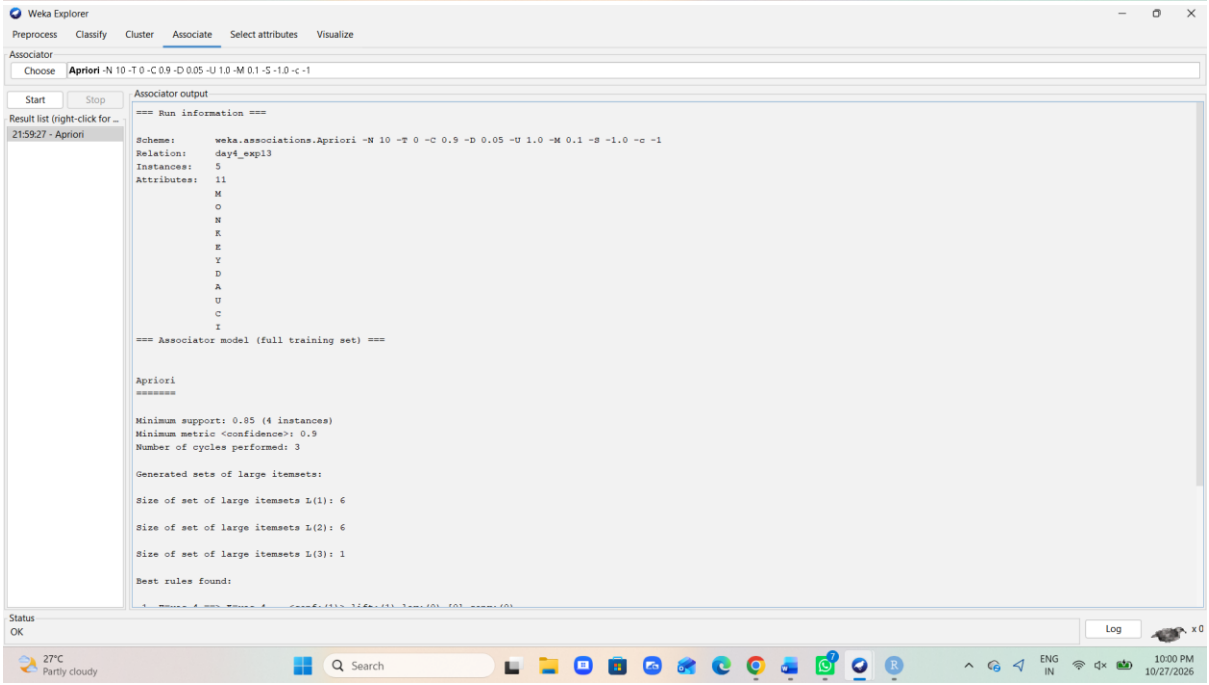
13.Consider this table

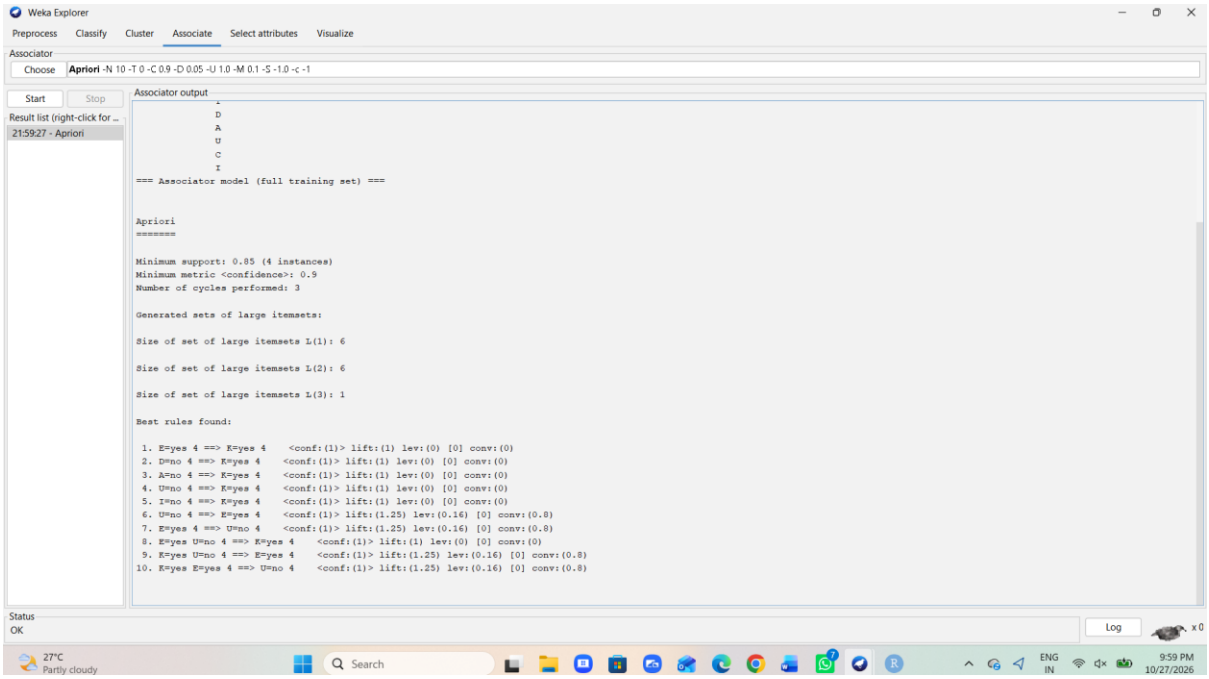
i) TID items bought
j) T100 {M, O, N, K, E, Y}
k) T200 {D, O, N, K, E, Y }
l) T300 {M, A, K, E}
m) T400 {M, U, C, K, Y}
n) **T500 {C, O, O, K, I ,E}**
o) (a) Find all frequent item set using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.
p) (b) List all of the strong association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers, and itemi denotes variables representing items (e.g., “A”, “B”, etc.):
q) ∀x ∈ transaction, buys(X, item1) ∧ buys(X, item2) ⇒ buys(X, item3)

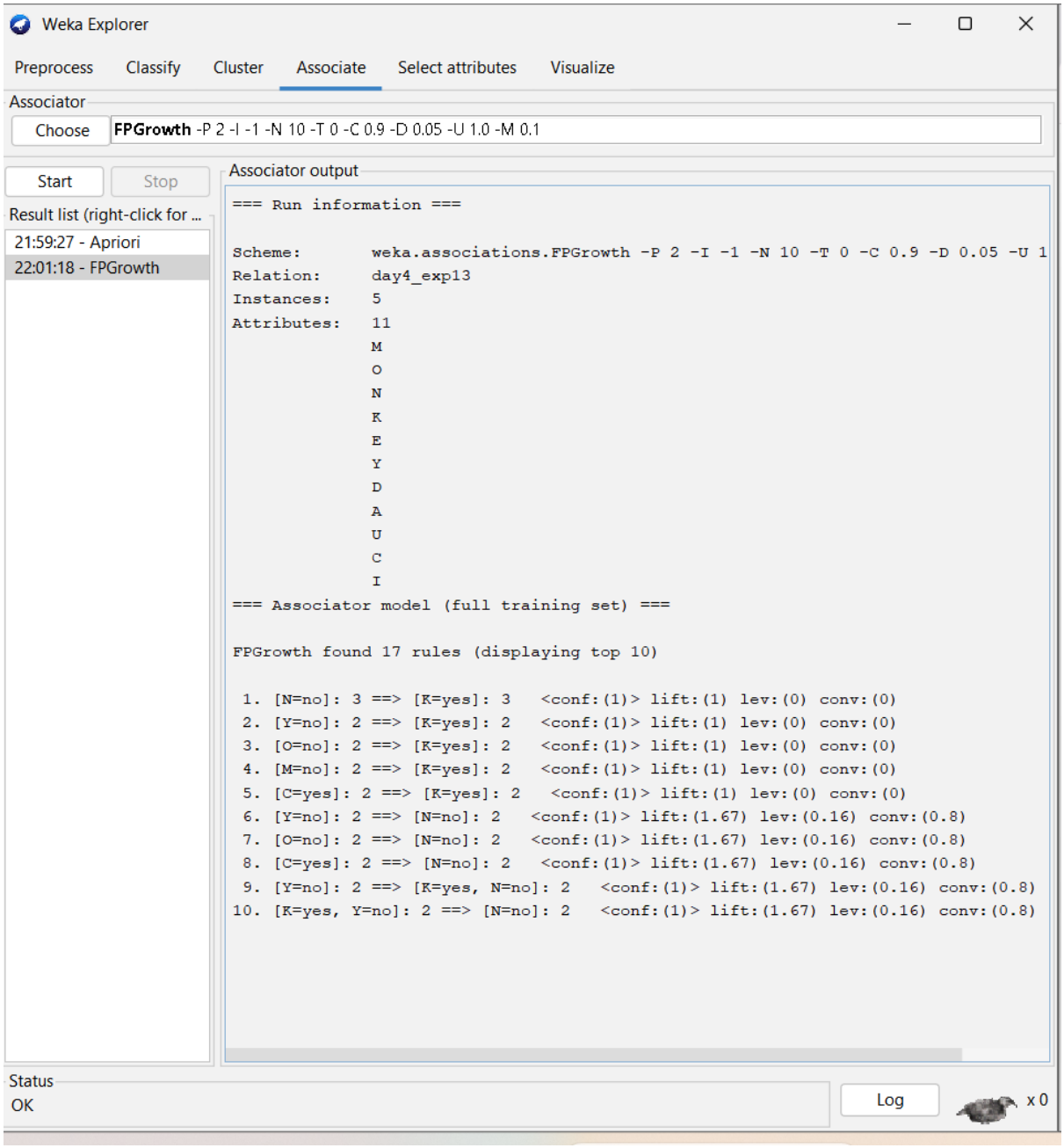
1. Aim

To find frequent itemsets and association rules using the Apriori and FP-Growth algorithms and compare their efficiency.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: day4_exp13
Instances: 5
Attributes: 11
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No.
Name
1 M
2 O
3 N
4 K
5 E
6 Y
7 D
8 A
9 U
10 C
11 I
Remove
Selected attribute
Name: M
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 0 (0%)
No.
Label
Count
Weight
1 yes 3 3
2 no 2 2
Class: I (Nom)
Visualize All
Status
OK
Log
x 0
27°C
Partly cloudy
Search
ENG
IN
9:58 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
21:59:27 - Apriori
Associator output
=== Run information ===
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Relation: day4_exp13
Instances: 5
Attributes: 11
M
O
N
K
E
Y
D
A
U
C
I
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.85 (4 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 3
Generated sets of large itemsets:
Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 6
Size of set of large itemsets L(3): 1
Best rules found:
Status
OK
Log
x 0
27°C
Partly cloudy
Search
ENG
IN
10:00 PM
10/27/2026


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
21:59:27 - Apriori
Associator output
I
D
A
U
C
I
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.85 (4 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 3
Generated sets of large itemsets:
Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 6
Size of set of large itemsets L(3): 1
Best rules found:
1. E=yes 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
2. D=no 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
3. A=no 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
4. U=no 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
5. I=no 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
6. U=no 4 ==> E=yes 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
7. E=yes 4 ==> U=no 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
8. E=yes U=no 4 ==> K=yes 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
9. K=yes U=no 4 ==> E=yes 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
10. K=yes E=yes 4 ==> U=no 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
Status
OK
Log
x 0
27°C
Partly cloudy
Search
ENG
IN
9:59 PM
10/27/2026



Result

Frequent itemsets and association rules were identified using Apriori and FP-Growth. Their execution time and generated rules can be compared in WEKA.